International Journal of Hydrogen Energy 141 (2025) 954–978

Contents lists available at ScienceDirect

International Journal of Hydrogen Energy

journal homepage: www.elsevier.com/locate/he

Sustainable energy prospects: Advancements in green hydrogen production through proton exchange membrane water electrolysis

Fazil Qureshi [a], Mohammad Asif [b], Abuzar Khan [c], Arafat Husain [a], Hamad Aldawsari [b], Mohammad Yusuf [d], Satish Kumar Verma [e], Mohd Yusuf Khan [c,f,*]

[a] Chemical and Petroleum Engineering Department, UAE University, P.O. Box 15551, Al Ain, United Arab Emirates
[b] Department of Finance College of Administrative and Financial Science Saudi Electronic University Riyadh 11673 Saudi Arabia
[c] Interdisciplinary Research Center for Hydrogen Technologies and Carbon Management (IRC-HTCM), King Fahd University of Petroleum & Minerals, Dhahran, 31261 Saudi Arabia
[d] Clean Energy Technologies Research Institute (CETRI), Faculty of Engineering and Applied Science, University of Regina, 3737 Wascana Parkway, Regina, SK, S4S 0A2, Canada
[e] Laboratory for Energy Storage Devices, Department of Physics, SSBSR, Sharda University, Greater Noida, India
[f] Materials Science and Engineering Department, King Fahd University of Petroleum & Minerals, Dhahran, 31261 Saudi Arabia

ARTICLE INFO

Handling Editor: Ibrahim Dincer

Keywords:
Proton exchange membrane water electrolysis (PEMWE)
Carbon circular economy
Sustainable green hydrogen
Electrolysis
Catalyst degradation

ABSTRACT

The imperative challenge for future socio-economic development lies in substituting fossil fuels with sustainable, environmentally friendly, and cost-effective energy sources and carriers. Hydrogen emerges as a frontrunner in this pursuit. Proton Exchange Membrane Water Electrolysis (PEMWE) stands out as a key technology for the sustainable generation of green hydrogen from water and intermittent renewable energy sources. The compact system design, high operating current density, elevated hydrogen purity, enhanced energy efficiency, and swift response, when integrated with renewable energy, make PEM water electrolysis particularly promising. This review provides a succinct overview of PEMWE technology, encompassing technical specifications, hydrogen market dynamics, and production costs, focusing on the commercial perspective. Additionally, we delve into the critical research surrounding PEM degradation mechanisms and strategies to augment PEMWE efficiency, essential pillars for technological progression. The review highlights notable research findings, addressing gaps and challenges that need to be addressed to overcome cost reduction and successful commercialization.

1. Introduction

The escalating global environmental challenges are exacerbated by a burgeoning population [1–3]. The surge in population contributes to heightened greenhouse gas emissions, driven by the expanding demands for both livestock husbandry raising and electrical energy, with the production causing carbon dioxide to be released (Fig. 1a). There is an expectation that energy demand will increase twofold by 2050 and triple by the end of the 21st century [4,5]. The swift exhaustion of fossil fuel reserves and the environmental consequences associated with their usage are major concerns for both policymakers and the general public. Consequently, there is an imperative need for a drastic transformation in the global energy consumption landscape over the coming decennium (Fig. 1b). In this context, water electrolysis emerges as a pivotal technology, not only offering pathways for green hydrogen production but also aligning with sustainability goals through its potential for real-world applications in energy storage and transportation. Scientists and engineers have a responsibility to seek out environmentally friendly energy sources derived from renewable resources with zero carbon emissions, such as wind, solar, hydroelectric, geothermal, and biomass energy [6–9].

As a promising electrochemical process, water electrolysis utilizes electrical energy to separate water (H_2O) into hydrogen (H_2) and oxygen (O_2). There are four primary types of water electrolysis: alkaline water electrolysis (AWE), anion exchange membrane water electrolysis (AEMWE), proton exchange membrane water electrolysis (PEMWE),

This article is part of a special issue entitled: ICHEAM-2024(Bhatnagar) published in International Journal of Hydrogen Energy.

* Corresponding author. Interdisciplinary Research Center for Hydrogen Technologies and Carbon Management (IRC-HTCM), King Fahd University of Petroleum & Minerals, Dhahran, 31261 Saudi Arabia.
E-mail address: mykhan@kfupm.edu.sa (M.Y. Khan).

https://doi.org/10.1016/j.ijhydene.2025.03.187
Received 15 September 2024; Received in revised form 3 March 2025; Accepted 12 March 2025
Available online 20 March 2025
0360-3199/© 2025 Hydrogen Energy Publications LLC. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

and solid oxide water electrolysis (SOEC). These electrolysis cells differ based on their electrolytes and ionic charge transfer (OH^-, H^+, O_2^-), although their fundamental working principles are similar [10–15]. Table 1 provides detailed technical specifications and a comparison of electrolysis technologies [16,17]. Different from AWE, PEMWE is compatible with common alterations in current load, which is vital for converting energy from renewable electricity sources. Each technology has positive and negative aspects, while the difficulties of lowering hydrogen production expenses differ depending on the technology [18]. For PEMWE, the membrane material significantly affects costs, making the development of affordable electrode materials crucial for cost reduction.

To achieve cost reduction, various challenges must be addressed, such as reducing the use of precious metals. Currently, the loadings of precious metals stand at 3.0 mg Ir per cm^2 and 1.0 mg Pt per cm^2, with the goal of decreasing them to 0.25 mg Ir per cm^2 and 0.05 mg Pt per cm^2 without compromising performance. Additionally, there is a need to replace precious metals with more economical transition metals [19,20]. Costly protective layers, such as platinum and gold, on bipolar plates should be minimized, and innovative designs using cost-effective materials should be explored (see section 5.2.4). Furthermore, reducing membrane thickness can significantly enhance cell efficiency. The current state-of-the-art membrane, Nafion® 117, has a thickness of approximately 180 μm, causing an efficiency (η) loss of about 25 % at an operating current density of 2 A cm^{-2} [21,22]. However, reducing the membrane thickness to as low as 20 μm is projected to result in an expected (η) loss of only 6 % at the same current density of 2 A cm^{-2} [23–25]. Researchers worldwide have been actively involved in developing advanced, low-cost PEM water electrolyzers, focusing on innovations in electrode materials and membranes.

To circumvent the need for costly water pre-treatment, contingent on country-specific regulations, electrolyzer technology needs to be adaptable to the naturally available water. However, the cost savings resulting from bypassing a purification step could be offset by diminished electrolyzer performance when utilizing impure water. Water from sources like the sea, lakes, rivers, and wastewater introduces challenging components that pose significant obstacles for both i.e., electrodes and membranes. Throughout this article, the authors consistently strive to pinpoint opportunities in various approaches, encompassing fundamental research areas like electrode development, material challenges, degradation mechanisms, and enhancing the (η) of the PEMWE approach. This review also delves into market considerations, cost-related issues, and the drivers behind cost-reduction efforts. The investigation of PEMWE systems' functioning materials and mechanisms has been the focus of numerous studies. However, in practical aspects, two key features of PEMWE systems have not been adequately

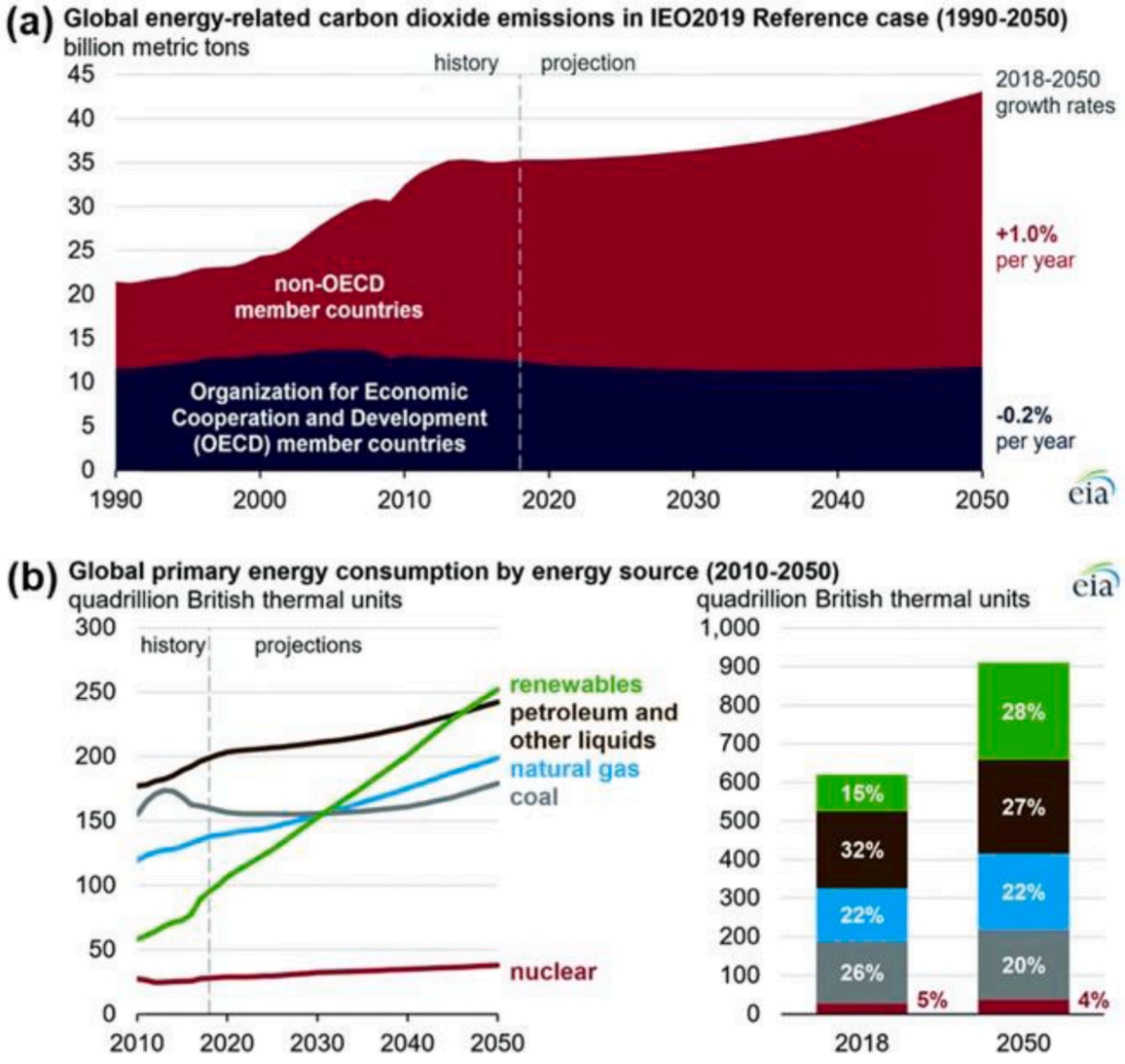


Fig: 1. (a) Global CO_2 emissions and (b) global consumption of primary energy by energy source adapted from [26,27].

Table 1
Comparison of technical specifications, advantages and disadvantages for various water electrolysis technologies [16,17,28].

Features	Alkaline	AEM	PEM	Solid oxide
TRL	Mature	R & D	Commercialized	R&D
Efficiency	50–78 %	57–59 %	50–83 %	89 % (lab)
Operating temperature	70–90 °C	40–60 °C	50–80 °C	700–850 °C
H_2 purity	99.5–99.9998 %	99.9–99.999 %	99.9–99.9999 %	99.90 %
Operating pressure	1–30 bar	<35 bar	<70 bar	<30 bar
Electrolyte	Alkaline solution (20 %–30 % KOH)	Solid polymer membrane with low concentrated alkaline solution (0–2 % KOH)	Solid polymer electrolyte (PFSA)	Solid ceramic separator with steam water
Electrode/catalyst (O_2 side)	Ni stainless steel	Ni/(NiFeCo) alloys	Pt-carbon	Perovskites (LSCF, LSM)
Electrode/catalyst (H_2 side)	Ni stainless steel	Ni	IrO_2	Ni/YSZ
Electrode area	>10000 cm^2	<300 cm^2	1500 cm^2	200 cm^2
Separator	Asbestos/Zirfon	Fumasep, Aemion	PFSA	YSZ
Durability	100000 h	30000 h	50000–100000 h	20000 h
Cathode PTL	Ni mesh/Porous Ni	Ni foam/carbon cloth	Carbon cloth/paper/STD	Ni foam/mesh
Anode GDL	Ni mesh/Porous Ni	Ni foam	Sintered titanium disk (STD)	Ni mesh/foam
BP	Stainless steel/nickel-coated stainless steel	Stainless steel/nickel- coated stainless steel	Platinum/gold-coated titanium or titanium	Cobalt-coated stainless steel
Separator plate	Ni-coated stainless steel	Ni-coated stainless steel	Gold/Pt-coated titanium	Cobalt-coated stainless steel
Nominal current density	0.2–0.8 A cm^{-2}	0.2–2 A cm^{-2}	1–2 A cm^{-2}	0.3–1 A cm^{-2}
Voltage range (limits)	1.4–2.5	1.4–2.0 V	1.4–2.5 V	1.0–2.0 V
Capital costs including BoP (1 MW)	US$ 640 per kW	US$ 542 per kW	US$ 587 per kW	US$ 2000 per kW
Capital costs including BoP (5 MW)	US$ 336 per kW	US$ 291 per kW	US$ 298 per kW	US$ 1500 per kW
Advantages	• Well commercialized technology • Long term stability • Megawatt range • Cost effective • Noble metal-free catalyst	• Non-noble metal catalyst • Low cost • Inexpensive polymeric membrane • High operating pressure (30 bar) • Small scale cell • Simple electrolyte or low concentration of alkaline solution • Compact cell design	• High current density • High gas purity • High voltage (η) • Significantly high operating pressure (up to 700 bars) • Dynamic operation • Good partial load range • Compact system design • Reduction of overvoltage • Rapid system response • Less energy consumption for compression	• Efficiency up to 100 % • The use of non-noble catalyst • Operatable under high pressure • Thermoneutral efficiency > 100 % with hot steam
Disadvantages	• Gas crossover (low purity) • Low operational pressure • Low dynamic operation • Low partial load range • Corrosive electrolyte • Low current density	• R&D stage • Poor durability • Membrane degradation • Low current density • Excessive catalyst loading	• High cost of components • Commercialized Stacks below MW range • Acidic corrosive environment • Possibly low durability • Reduction of stack component lifetime • Reduction of current (η) • Susceptible to increased potential difference equilibrium • Electrolyzer safety problem	• R&D stage • Low durability (brittle ceramic) • Bulky system design • Non-dependable cost information

surveyed and estimated:

(i) The feasibility and reliability of PEMWE systems that are proposed in the lab must be assessed under real conditions.
(ii) The disparity between the performance of cutting-edge PEMWEs in the literature and the demands of real-world applications.

This review aims to address these gaps highlighting the recent advances in PEMWE technology. It mainly focuses on providing a comprehensive assessment of the individual components in PEMWEs and proposes a more pragmatic evaluation by considering the entire cell configuration and working protocols. Analogously to PEM fuel cells, any operation causes the degradation of one or more components. It provides a detailed overview of the degradation mechanisms and outlines corresponding mitigation strategies for components such as catalysts, membranes, PTL, and BPPs. Additionally, this review also identifies research gaps that need to be addressed to overcome freshwater supply challenges for hydrogen production. Furthermore, the review also analyzes the operating conditions, discusses the main technical challenges associated with stack components for durable operation, and introduces recent approaches to overcome these technical challenges. To bridge the gap between lab accomplishments and real-world situations, it focuses on benchmarking materials that have been widely utilized in the PEMWE system and also discusses strategies for reducing expensive materials usage and economic effectiveness. Following an overall understanding of this system, we finally provide perspectives on the remaining challenges facing the future study of PEMWEs. This timely review aims to offer a comprehensive and practical assessment of the current achievements and challenges, aiding in the further development of affordable PEMWEs.

2. Water electrolysis for hydrogen production

Water electrolysis (WE), unlike the other technologies, does not rely on fossil fuels and produces no GHG emissions during the process. WE do not utilize any carbon feed directly. Therefore, H_2 generated through WE is commonly referred to as carbon-free or green H_2 [29–31]. WE are the electrochemical water splitting method for green H_2 with the help of electricity, and the basic reaction of WE require 1.23 V theoretical thermodynamic cell voltage that splits water into H_2 and O_2 at room temperature (Eqn (1)). Nonetheless, plant generation, infrastructure development, and the underlying water and electricity supply chains

could all have a significant impact on climate change.

$$H_2O \rightleftarrows 0.5\ O_2 + H_2\ \Delta H_R^0 = +285.9\ kJ/\ mol \quad (1)$$

WE is based on an ion-conducting electrolyte and two electrodes, which make-up an electrolytic cell (EC) [32]. Electricity is supplied in the form of direct current to create an electric potential difference, or voltage, among the electrodes. An EC experiences both a non-spontaneous ($\Delta G_R>0$) and endothermic ($\Delta H_R>0$) redox reaction when an elevated voltage is supplied. Deionized H_2O undergoes electrochemical breakdown at one of the electrodes, depending on the specific kind of electrolyzer, producing either H_2 and negatively charged ions (anions) through reduction or O_2 and positively charged ions (cations) through oxidation. Attracted by the opposing electrode's charge, the ions subsequently move towards it. The ions that the electrodes attract are the reason behind their names. Anions migrate to the anode, while cations migrate to the cathode. The next stage of the redox reaction occurs at the second electrode, where H_2 and O_2 are produced as a result of reduction and oxidation at the cathode and anode, respectively. Equation (1) shows that the H_2O splitting enthalpy of the reaction (ΔH_R) is divided into two parts: thermal (ΔS_R) and electrical (ΔG_R) [33,34]. According to the primary rule of thermodynamics, an alteration in Gibbs free energy equals the amount of energy required to electrochemically break down one molecule of H_2O (eq. (2)). As a result of this, the Faraday constant (F = 96, 485 °C/mol), which measures the charge of electricity per electron mole, and the electrons transferred per mole of broken-down H_2O (n = 2) may be used to calculate the theoretically needed voltage (U_{th}) of a WE cell (eq. (2)) [33,34].

$$\Delta H_R = \Delta G_R + \Delta S_R \times T \quad (2)$$

$$\Delta G_R = U_{th} \times n \times F \quad (3)$$

Water splitting at standard pressure (1 bar) and temperature (25 °C) results in the following thermodynamic properties [23,35]:

$\Delta G_R^0 = 237.21$ kJ/mol,

$\Delta S_R^0 = 0.1631$ kJ/(mol·K),

and $\Delta H_R^0 = 285.84$ kJ/mol.

If all of the required process heat is supplied externally, the minimum reversible voltage of a cell for H_2O splitting is 1.23 V. Electricity is commonly used to supply both thermal and electrical energy, including the ΔH_R^0, resulting in a 1.48 V cell voltage. It ought to be emphasized that in practice, ohmic resistance, mass transportation restrictions, and losses of activation at the electrodes all contribute to a cell's excess potential, which results in a much higher actual voltage of the cell (U_{cell}). This results in a drop in the WE cell's voltage efficiency (ηV), which is determined by dividing the actual voltage of the cell by the theoretical voltage of the cell, as shown by Eq. (4).

$$\eta V = U_{th}/U_{cell} \quad (4)$$

To determine the voltage efficiency, the combined state's HHV (Higher Heating Value) or LHV (Lower Heating Value) is employed, depending on the type of H_2O provided. In the case of liquid water, the HHV of water is the relevant metric. However, for steam-powered electrolyzers, the LHV is more appropriate as the water is already vaporized prior to entering the electrolyzer, eliminating the need for energy expenditure on evaporation. It is also important to distinguish whether the generated H_2 is used chemically HHV or energetically LHV, as well as whether the O_2 by-product is used further. One alternative method to determine the (η) of an electrolyzer is to divide the amount of energy of H_2 by its specific consumption of electricity needed to generate 1 Nm^3 H_2 under typical circumstances (W_{el}), as determined by its LHV (3.00 kWh/Nm^3) or HHV (3.54 kWh/Nm^3). The total energy consumption of all the process devices is incorporated in when calculating the H_2 generation system's (η) (eq. (5)).

$$\eta_{LHV} = (LHV * \eta_{HHV})/W_{el} = HHV/W_{el} \quad (5)$$

AWE, PEMWE, and high-temperature SOEC are the three most common WE technologies for producing H_2. The development stage, the charge carrier and electrolyte, standard operational conditions, and the appropriateness of the constant operation arising from an interrupted supply of power from renewable power sources were the main differences among the electrolysis technologies. The most cutting-edge and practically useful WE technique on the market presently is AWE. Despite its unique capability for dynamic operation dependent on the accessibility of power provided by intermittent renewable energy (RE) sources, PEMWE is the subject of this study. High-density currents, compact designs for systems, safe elevated-pressure operation that eliminates the need for further compression, energy-effective ready operation, quick response, shut-down, start-up times, a wide partial load range, and a high purity of product gas are some of PEMEL's other promising characteristics.

A common PEMWE cell consists of the following components: (1) a solid polymer electrolyte; (2) two electrodes with catalysts composed of precious metals, the cathode and anode; (3) two porous transport layers (PTL); and (4) two bipolar plates (BP), as seen in Fig. 2. The electrolyte is typically Nafion™ or fumapem™, a highly porous polymeric membrane that conducts protons. The membrane's porosity (thickness 60–200 μm) [36] improves gas and H_2O movement as well as the flow of current. To reduce ohmic resistances, electrodes (thickness 10 μm) are directly deposited on the membrane. The membrane is covered with noble metal catalysts, which are typically based on Ir (anode) and Pt (cathode). Carbon paper (cathode) and sintered Ti foam or felt (anode) make up the surrounding PTLs (thickness: 280 μm. The PTLs are securely attached to BPs, which are typically made of Ti. Transport of H_2O and product gases is made possible by the carved channels for flow on the plates. Furthermore, the channels provide a uniform dispersion of current density, ensuring that the whole region is utilized uniformly for the separation of H_2O. Electrically connected in series, multiple electrolysis cells form a single electrolyzer stack. Due to their resilience to high anode potential and the corrosive, acidic environment created by the proton carrying ionomers, platinum, iridium, and titanium are essential, scarce, and expensive materials. Increasing current densities, decreasing catalyst loading, and conserving critical materials are the three main research objectives at this time.

PEMWE involves feeding liquid, deionizing H_2O on the anode side, then applying voltage to induce the H_2O to oxidize into O_2 and protons (eq. (6)). As a result, gaseous O_2 is released from the system. Once at the cathode, protons cross the membrane and combine with electrons to generate H_2 (eq. (7)), which is then removed in the form of gas.

Reaction at anode:

$$H_2O \rightleftharpoons 0.5\ O_2 + 2\ H^+ + 2\ e^- \quad (6)$$

Reaction at cathode:

$$2\ H^+ + 2\ e^- \rightleftharpoons H_2 \quad (7)$$

PEMWE normally operates at pressures of up to 200 bar and temperatures of 20–150 °C. Electrolyzers operating at pressures greater than 30 bars have been merely studied in small-scale experiments so far. PEMWE stacks have a voltage (η) of 67–82 %, with cell voltages of 1.4–2.2 V and current density of up to 4.0 A/cm^2. Average to small-scale PEMWE electrolyzers are available to be purchased commercially (TRL 9). In 2020, PEMWE began commercial operation of the globe's largest green H_2 generation plant, which is based on four 5 MW electrolyzers in Quebec, Canada [37]. With a 20 MW capacity, the plant generates around 3000 t/a of H_2 annually. Both H_2 and O_2 are generated with excellent purity and may be utilized with no additional processing. PEMWEs have several benefits over alternative water electrolyzers, as shown in Table 1: (1) the ability to operate at high current levels (strong strength); (2) high power productivity; (3) high gas purity; and (4) a

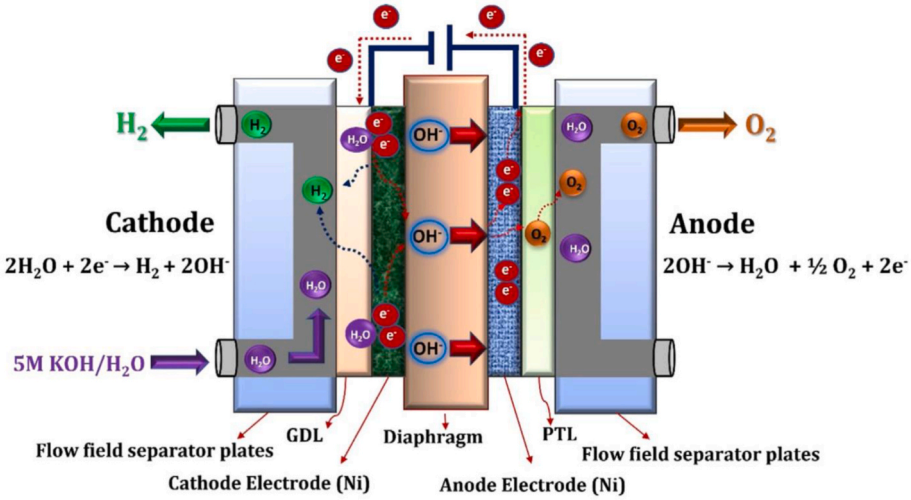


Fig: 2. Illustration of a PEMWE cell's adapted from [17].

wide energetic array, making them perfect to be utilized in periodic sustainable energy. However, PEMWEs have some disadvantages, including (1) a significant start-up expense and (2) the need for elevated temperature electrolyzers.

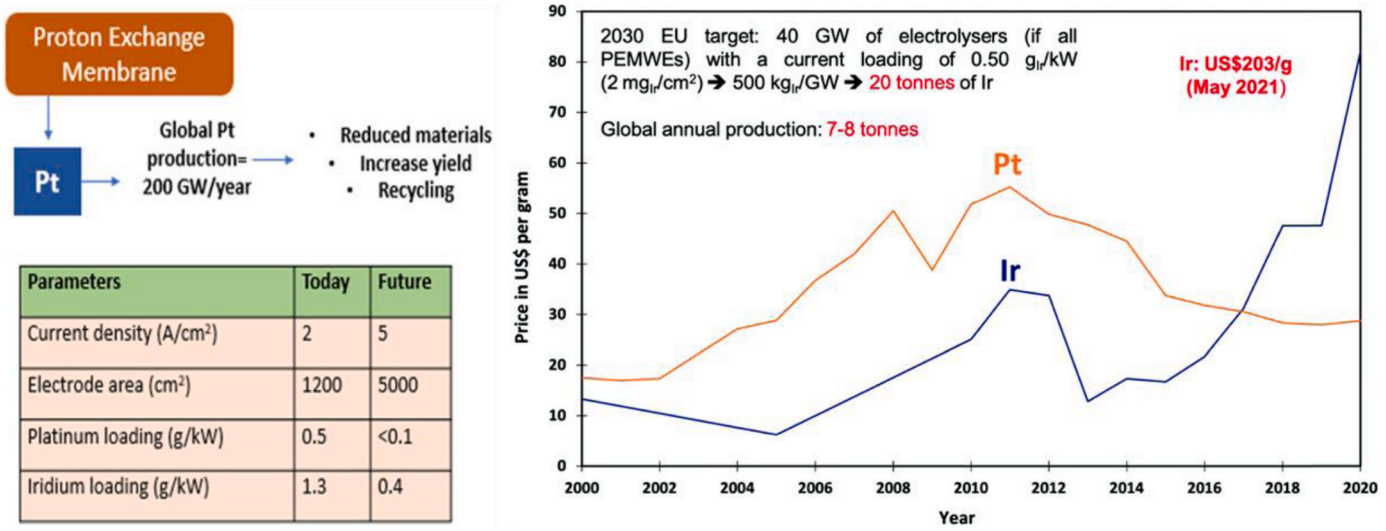
3. PEMWE materials: focus, challenges, and solutions

Green hydrogen production, which is achieved by integrating renewable energy structures with electrolysis facilities, is experiencing a notable surge. According to the Hydrogen Council, hydrogen could meet approximately 25 % of the world's energy needs by 2050, establishing a $10 trillion market. This positive outlook is reinforced by robust national hydrogen strategies in various EU countries and Asian countries. Additionally, Aurora Energy Research forecasts a significant 1000-times rise in electrolyzer components by 2040 [38,39]. While the majority of environmentally friendly hydrogen ventures currently favor well-established technologies such as PEMWEs and AWEs, new innovations such as AEMWE and SOEC are being investigated for extensive hydrogen generation. Water electrolysis, the principal electrochemical method for generating hydrogen, is predicted to become increasingly important as renewable energy grows [40,41]. However, the effectiveness and affordability of water electrolysis technologies are determined by a variety of factors, including materials (catalysts, electrolytes, separators) and operating conditions (temperatures and pressures). Electrolytic hydrogen generation is currently more expensive than other technologies due to substantial upfront expenses and reliance on power prices. Despite decreases in Capital Expenditure (CAPEX) and Operational Expenditure (OPEX) seen since 2012, Despite decreases in CAPEX and OPEX since 2012, more progress is needed, especially when relying primarily on sustainable energy. To achieve commercial viability, electrolyzer systems must be both cost-effective and efficient. As a general rule, all electrolysis methods require the creation of new substances that are economical, high-performing, and long-lasting. Thinner membranes (electrolytes), more effective catalysts, and the use of less crucial raw materials are all priorities in this pursuit of improvement [26].

As previously stated, PEMWEs rely on costly and rare elements including iridium (Ir) and platinum (Pt) for the two electrodes, along with titanium-based components in the permeable transport membrane. Platinum Group Metals (PGMs) are currently being loaded at the anode at 2–5 mg IrO_2 cm^2 and the cathode at 1–2 mg Pt cm^2. For a 100 MW PEMWE working at 4 W cm^2, this amounts to approximately 50 kg of Ir, evaluated at the price of US$203 per gram, culminating in an astounding cost of US$10.15 million. Furthermore, the shortage of Ir and Pt in the planet's core exacerbates the issues. As reported by Minke et al. [42], the yearly production rates of iridium and platinum are approximately 7–8 and 200 tons, respectively, with major mining operations in Canada, Russia, South Africa, the United States of America, and Zimbabwe. South Africa stands out as the main producer, accounting for 70 % of the world's resources. The report warns of future iridium bottleneck availability if loading is not greatly decreased and PGMs are not fully recycled, particularly with installed capacity expected to rise at an average rate of 2 GW $year^{-1}$ over the next 50 years. For example, if the EU's objective of 40 GW of electrolyzers by 2030 is based primarily on PEMWEs with a current loading of 0.50 g Ir kW^{-1}, roughly 20 tons of iridium will be necessary. Addressing PGM volatility in prices and shortage is a crucial target, intending to reduce their load by at least 40, notably for iridium, and eventually replace them with PGM-free catalysts. This reduction is attained through the advancement of non-carbon-based catalysts with high surface areas (HSA), blending precious group metals (PGMs) with readily available materials such as other transition metals, enhancing catalyst surface area using improved manufacturing techniques, and refining catalyst utilization within membrane electrode assemblies using improved dispersion and deposition methods. Fig. 3 depicts the progression of platinum and iridium costs (US$ g^{-1}) from 2000 to 2020, highlighting a shift in which iridium has turned costlier than platinum since 2017, with a staggering 20-fold spike in price since 2013.

3.1. Access to water: an additional challenge

Large-scale "green" hydrogen production implicates the use of vast amounts of freshwater [43] primarily collected in groundwater systems, lakes, and rivers, jointly representing ca. 0.77 % of the total water estimated for the Earth's shallow reservoirs (1386*106 km^3). In general, it is necessary to use 9 kg of freshwater (roughly equivalent to 9 L) to produce 1 kg of hydrogen, which highlights the vast amounts of water needed and the possibility that hydrogen production will compete with other usages of freshwater and increase the pressure on this highly vulnerable natural resource. Although the overall use of water in a hydrogen economy is inferior to thermal, nuclear, and coal plants, water consumption reduction may be offset if proper precautions are not taken [44]. Access to freshwater may be a relevant challenge in some locations with imminent (or forecasted) shortages in freshwater, which should not be ignored when planning the implementation of infrastructures for large-scale "green" hydrogen production. Moreover, the shortages will likely worsen due to pollution, climate change, overpopulation, and inefficient use of resources.



Parameters	Today	Future
Current density (A/cm²)	2	5
Electrode area (cm²)	1200	5000
Platinum loading (g/kW)	0.5	<0.1
Iridium loading (g/kW)	1.3	0.4

Fig: 3. The price trend of iridium and platinum (US$ g^{-1}) between 2000 and 2020 adapted and modified from [26].

Four approaches need consideration to circumvent the problem of using freshwater resources: i) use of wastewater from water treatment centers; ii) use of water from desalinization plants; iii) direct use of seawater for hydrogen production; and iv) use of local rainwater. The use of wastewater will likely increase the cost of hydrogen as the water treatment centers are dispersed, and it needs to be transported to the hydrogen hubs [45]. Alternatively, a decentralized model for hydrogen production, with small production hubs close to treatment centers, would solve the transportation needs. Seawater represents ca. 96.54 % of the Earth's shallow reservoirs, and desalinization technologies are well-known. There are, however, significant challenges with these technologies: (i) expensive running costs; (ii) energy-intensive and often fossil-fuelled; and (iii) produced wastes (brines) must be disposed of safely [46–49]. To lessen GHG emissions, desalinization plants would need to run on renewable energy, and today, its share is below 5 %. The energy (η) and the reduction of brines, or their use as a source of chemical elements, are also gaining momentum [50,51] but are still far from being implemented worldwide in the short term. Using water desalinization would likely increase the cost of hydrogen and its environmental impacts.

The direct use of seawater to generate hydrogen is possible, as different technologies exist [52–54], but still with a low readiness level and hardly feasible/affordable in the coming years [55]. Furthermore, according to the proposed solutions for catalysts and other components, the material requirements look quite significant, and there are no evident advances on how to deal with the resulting wastes, just like with the desalinization. Using local rainwater would require the construction of water retainment structures and proper studies of precipitation at a local level. Juarez-Casildo et al. [56] estimate that the use of 5.3 % of precipitation at the national level would satisfy the Mexican need for hydrogen, but detailed studies at the local level found areas where more than 100 % of the precipitation would be needed. A high-resolution study at the local level is necessary to assess the resource correctly, but the use of rainwater may help to curb the use of freshwater resources significantly. Water use is frequently neglected in hydrogen analysis [57] but is a vital stock to the functioning of electrolyzers. It is necessary to develop management strategies for the water resource, optimize its use, and avoid waste, and possible recycling techniques. Ensuring system designs account for water consumption can foster appropriate deployment of hydrogen technologies. Table 2 shows the impurity effects categorized by source and effects on PEMWE components.

Seawater, brackish water, wastewater, and rainwater, among other things, are viewed as promising resources for meeting freshwater demand in hydrogen production. Freshwater has a total dissolved solid (TDS) of less than 0.5 g/kg, while seawater, wastewater, rainwater, and brackish water have TDSs of 35–45 g/kg, 0.5–2 g/kg, 0.01–0.15 g/kg, and 1–35 g/kg, respectively [58,59]. The desalination method can be used to remove salt and minerals from seawater or brackish water,

Table 2
Impurity effects categorized by source and effects on PEMWE components adapted from [89].

Impurity	Source	Cell components	Membrane	Anode	Cathode
Ca^{2+}, Na^{+}, Mg^{2+}	Seawater, potable water External contamination		Lowers membrane conductivity Lowers membrane conductivity, degrades membrane	Lowers ionomer conductivity, catalyst poison	Lowers ionomer conductivity, decreases HER rate
Ni^{2+}, Cu^{2+} Fe^{2+}, Fe^{3+} Ti^{4+}	Stainless steel tubing degradation PTL corrosion				
Ir^{n+} Cl^{-}	Anode catalyst dissolution Water purification failure			Enhances dissolution, unwanted side product, poison catalyst	
F^{-}	Water purification failure, membrane degradation	Corrodes Ti PTL and BPP		Enhances dissolution	
I^{-}, Br^{-}	Water purification failure			Catalyst poison, enhances dissolution	
SO_4^{2-}	Water purification failure, membrane degradation			Catalyst poison	
Organics	External contamination			Catalyst poison, enhances dissolution, unwanted side product	
Solid particulates	External contamination			Blocks PTL pores and induces mass transport issues	

allowing it to be used as an electrolysis feedstock. Both the electrolysis process and the necessary feed water purification consume energy, but the electrolysis process consumes significantly more energy than the water treatment stage. The most significant challenge for direct seawater electrolysis is complex catalysis and catalyst working (η) due to competitive chlorine evolution reaction (CER) at anode, which is corrosive to the system, and metal deposition at the cathode. During the CER, chlorine gas is produced at the anode according to equation (8), competing with the anodic oxygen evolution reaction (OER, equation (9)) [60]. CER, which involves the transfer of 2e-as opposed to 4e-transfer in OER, has faster reaction kinetics and can degrade the activity and stability of the OER catalyst as electrolysis progresses. CER, on the other hand, can be suppressed in alkaline electrolytes due to OER's thermodynamic preference. Still, hypochlorite formation can occur at potentials greater than 490 mV, which must be addressed to realize commercial applications [61]. Various modification strategies, such as protective layer coating, alloying, and heteroatom doping, have previously been used to mitigate the issue of CER. In a recent study, a transition metal oxide was coated with a Lewis acid layer to manipulate the local reaction environment for direct seawater splitting by capturing the hydroxyl anions, which improves reaction kinetics while preventing corrosive chlorine chemistry at the anode [54]. Another approach was to synthesize a transition metal phosphide with a polyanion layer on the catalyst's surface containing phosphate and nitrate anions, which helps to repel chloride anions from attacking the catalyst's surface, thereby preventing corrosive chlorine chemistry.

$$2Cl- \rightarrow Cl_2 + 2e^-,\ U^\circ_{CER} = 1.36\ V\ vs.\ SHE\ (Standard\ Hydrogen\ Electrode) \quad (8)$$

$$2H_2O \rightarrow O_2 + 4H^- + 4e^-,\ U^\circ_{OER} = (1.23 - 0.059 \times pH)\ V\ vs.\ SHE \quad (9)$$

The thermodynamics of seawater electrolysis favor alkaline conditions because the theoretical potential for CER and hypochlorite/chlorate formation is greater than that of OER at the anode [62,63]. Although operating under stronger alkaline conditions may help to reduce the CER, it additionally raises the possibility of $Ca(OH)_2$, $Mg(OH)_2$, and other hydroxide depositions on the cathode side [64]. The presence of Mg_2^+, Ca_2^+, and Na^+ in seawater can reduce the electrocatalytic activity and stability of hydrogen evolution reactions. Moreover, as current densities rise, the pH at the electrode-electrolyte interface rises, allowing more hydroxide to accumulate on the electrode's surface, and blocking active sites on the catalyst's surface. Using buffer solutions to stabilize the pH of the electrolyte at the cathodic end can be a beneficial strategy. Buffer solutions can prevent unwanted pH fluctuations in the electrolyte, thereby preventing hydroxide accumulation from blocking active sites. Furthermore, the pH level can be maintained by improving the design of an electrolysis system that can detect sediment attachment to the cathode and remove it from the cathode. However, adding a buffer to the feed can dramatically increase the cost. Furthermore, during electrolysis, the buffer solutions can remain in the electrolyzer for a long time, resulting in an increased concentration of seawater until precipitation [55]. These issues can be addressed with solid electrolyte membranes, but more research is needed to realize their application because these membranes are sensitive to impurities that can potentially block ion exchange sites, resulting in membrane deactivation and high maintenance costs. The addition of buffers, particularly alkaline buffers, can cause precipitation of Mg and Ca ions in seawater, and researchers have suggested centrifugation and filtration prior to electrolysis as a way to eliminate these precipitates [65]. A lot of effort has gone into developing membranes to block unwanted anions during direct seawater splitting; however, these membranes degrade over time. Maintaining a high level of hydration in the membrane is an effective way to reduce membrane degradation, which is especially important when the seawater is highly impure. Nonetheless, significant effort is required to develop an efficient and effective method for removing precipitate and salt build-up during continuous operation.

Recently, a variety of strategies have been used to develop efficient catalysts for direct seawater electrolysis, including the use of simulated alkaline seawater and real seawater. Yu et al. [62] developed a self-supported nickel-iron phosphosulfide (NiFeSP) nanotube array electrode for direct seawater electrolysis using a two-step sulfurization/phosphorization method. To achieve a current density of 0.5 A/cm^2 in a simulated alkaline-seawater solution, the as-synthesized catalyst required an overpotential of 0.38 V for hydrogen evolution reaction (HER) and 0.26 V for OER due to under-coordinated metal sites and abundant heterointerfaces. As employed as a bifunctional catalyst, it achieved 0.5 A/cm^2 at 1.938 V and remained stable for more than 1000 h with no noticeable performance degradation. Guo et al. [54] compared the seawater splitting performance of CoOx, NiMo–Fe-doped NiOOH, and Pt/C–RuO_2 by adding a Lewis acid layer (Cr_2O_3) to the surface. By coating CoOx with Cr_2O_3, in-situ hydroxyl anions were generated around the catalyst, which aided in the splitting of water molecules. These in-situ generated hydroxyl anions were responsible for the formation of local alkalinity around the catalyst while also repelling chloride anions in the seawater, preventing CER and electrode corrosion. There have been numerous reports of using various types of polyanion coatings to prevent chlorine chemistry at the anode with reasonable success, but they have yet to be adopted for industrial applications [66]. In reacting, Xie et al. [67] devised a method for producing hydrogen using direct seawater electrolysis that can address side reactions and corrosion issues. An in-situ water purification process was combined with the water electrolysis system using a self-driven phase transition mechanism. As a gas-path interface, a waterproof breathable membrane made of hydrophobic porous polytetrafluoroethylene (PTFE) was used, with concentrated potassium hydroxide (KOH) solution serving as a self-dampening electrolyte. The membrane prevented liquid seawater and impurity ions from penetrating while allowing water vapor to diffuse. As a result, the system remained stable for more than 3200 h under industrial conditions at a current density of 0.25 A/cm^2. Loomba et al. [68] synthesized a unique catalyst composed of porous sheets of nitrogen-doped $NiMo_3P$, where N and Ni assisted in the creation of new active sites and improvement of electrical conductivity by tuning the electron density of Mo, the metal-non-metal bond lengths, and optimizing the surface energies, and compared it to other catalyst performance. The existence of M − N bonds and polyanions (phosphate and nitrate) on the surface protected the catalyst from chlorine chemistry and improved stability. As a result, N–$NiMo_3P$ required only 23 and 35 mV overpotentials to achieve 10 mA/cm^2 for hydrogen evolution reactions in KOH and alkaline seawater, and 1.52 V and 1.55 V for full water splitting to achieve 10 mA/cm^2 in KOH and seawater, respectively. It has been a significant challenge to develop catalysts for direct seawater splitting at a commercial scale in comparison to freshwater, but recent progress indicates that we are on the right track, to commercialize these catalysts shortly.

Considering possibilities for commercialization of direct seawater electrolysis, the complex nature of seawater, which contains a variety of minerals, salts, suspended pollutants, and biological species, suggests that challenges must be fully addressed. On the other hand, seawater desalination is proposed as an alternative solution, but the key factors to consider are the mismatch in technology scale, the need for capital investment, and operational costs. Several desalination techniques have been developed and classified into different types based on the method used to generate freshwater [69–71]. Seawater desalination technology advancements demonstrate the potential to reduce the current freshwater deficit for hydrogen production at a low cost, particularly in coastal areas [72]. For example, scientists have reduced the specific energy requirement of desalination from 20 $kWhe/m^3$ in 1970 to 2.5 $kWhe/m^3$ in 2021 for seawater reverse osmosis (RO) desalination. Temiz and Dincer [73] investigated a renewable integrated system that produces freshwater via multi-effect distillation and hydrogen via a PEM electrolyzer. They determined that their system could produce 402 tons

of water at 46 kg/h and 283 tons of hydrogen at 32.32 kg/h.

To produce hydrogen, electrolyzers require high-purity water that meets ASTM standards type II or type I [74]. Although seawater RO (SWRO) can provide potable freshwater at a much lower cost than membrane distillation (MD) and multi-effect distillation (MED), the supplied water from RO contains TDS of about 0.6 g/kg, necessitating the addition of passes and ion exchange or electro-deionization (EDI) techniques to demineralize the water product from the RO, making it suitable for electrolysis, which increases the overall cost [75]. The expenditures on capital of an RO unit and an ion exchange resin unit for producing clean freshwater from RO at a rate of 100 m^3/h are $264300 and $187238, respectively. The operating and maintenance costs of the ion exchange resin unit are $13000 per year while assuming that resin replacement is required every 2–3 years, the replacement cost per kg resin would be $5.44 [76]. RO, multi-stage flash desalination (MSF), MED, and membrane distillation are currently the most commonly used methods, accounting for nearly 69 %, 17 %, 7 %, and 1 % of the total desalination market capacity, correspondingly [77]. While RO is a successful water desalination technique, it operates at high pressure, which is frequently associated with high energy demand and relies solely on electricity, similar to electrodialysis (ED) [78]. MD outperforms RO in terms of salt rejection, producing nearly 99 % of the water for large-scale freshwater production while using low-grade thermal energy. However, because of the high costs of producing freshwater, it trails RO for small and medium-sized applications. For large-scale applications, they have comparable water production costs [79,80].

Each year, 380 billion m^3 of wastewater is produced around the world. Asia produces 42 % (159 billion m^3) of this wastewater, while Europe and North America generate 68 billion m3 and 67 billion m3, respectively [81]. Wastewater production is projected to rise by 24 % and 51 % by 2030 and 2050, respectively [82]. This suggests that these wastewater resources could be used to produce hydrogen [83–85]. Cvetkovi'c et al. [86] studied hydrogen production by converting food industry wastewater into biogas. The research, which focused on the Serbian food industry, found that two technological methods, biomass steam reforming, and electrolysis after biomass cogeneration, are suitable for H_2 generation. Their findings show that biomass steam reforming generates nearly 4095 tons of hydrogen per year, whereas electrolysis generates slightly more than 1751 tons. Qyyum et al. [87] investigated a variety of methods, including fermentation, sequential dark, and photo-fermentation. To generate hydrogen from a variety of industrial wastewater, including olive mill wastewater, sugar beetroot wastewater, dairy wastewater, textile wastewater, rice mill wastewater, and so on. To ensure sustainability. Chaurasia and Mondal [88] use microbial electrolysis cells for wastewater treatment and hydrogen production in the sugar industry. They discovered that H_2 can be produced at a rate of 0.24 m^3/day with wastewater treatment (η) of 50 %.

4. PEMWE degradation mechanism

In comparison to alternative methods of hydrogen production, PEM water electrolysis technology needs to extend its operational duration to more than 10^{-4}–10^{-5} h to attain economic competitiveness [90]. To achieve this objective, a deeper understanding of the microscopic processes that contribute to the long-term decline in electrolyzer performance is essential, along with the identification of effective solutions. Over the past decade, researchers have dedicated significant efforts to unraveling the degradation mechanisms in PEM fuel cell [91]. However, there is a notable scarcity of reports addressing the degradation of PEMWEs. Despite some similarities between the two technologies, their differences are substantial. Particularly, the intermittent and variable operating conditions inherent in PEMWEs tend to significantly elevate the degradation rate [92].

4.1. Degradation mechanisms of catalyst layer

Catalyst stability and durability in PEMWE environments are challenged by factors such as high anodic overpotential, strong oxidizing situations, low pH, and elevated temperatures. These challenges can cause catalyst dissolution, metal ion blocking of proton exchange sites, agglomeration, and support passivation/corrosion. Such issues cause poor contact between the catalyst layer (CL) and the electrodes, as well as altered interfaces between the membrane and the CL, lowering the ECSA while increasing electrical resistance.

4.1.1. Catalyst dissolution/agglomeration

Operating for a long time at high anodic potential (41.6 V) and low pH can result in Ir catalyst dissolution (Fig. 4a and b) [93]. The dissolution mechanism, involving intermediates like IrO_2OH and $HIrO_2$ (Eqn (10)–(16)), is not fully agreed upon in the literature.

$$HIrO_2 + 3H^+ \rightarrow Ir_3^+ + 2H_2O \quad (10)$$

$$HIrO_2 \rightarrow IrO_2 + H^+ + e^- \quad (11)$$

$$IrO_2 + H_2O \rightarrow IrO_2OH + H^+ + e^- \quad (12)$$

$$2IrO_2OH \rightarrow 2HIrO_2 + O_2 \uparrow \quad (13)$$

$$IrO_2OH \rightarrow IrO_3 + H^+ + e^- \quad (14)$$

$$IrO_3 + H_2O \rightarrow IrO_4^{2-} + 2H^+ \quad (15)$$

$$2IrO_3 \rightarrow 2IrO_2 + O_2 \uparrow \quad (16)$$

Specifically, Eqn (10) describes the dissolution of the iridium intermediate, $HIrO_2$, under acidic conditions. H^+ facilitates the oxidation of $HIrO_2$, producing Ir_3^+ and water. This breakdown leads to a reduction of active Ir species, which may decrease catalyst durability under long-term high-potential operation. The existence of Ir_3^+ suggests that iridium is leachable in the acidic electrolyte. In the following oxidation reaction, $HIrO_2$ is converted to IrO_2, releasing H^+ and e−. IrO_2 tends to be more stable under typical OER conditions, but this transformation marks a step in which additional oxidation can occur, contributing to higher oxidative states and potential catalyst degradation (Eqn (11)). In Eqn (12), IrO_2 interacts with water to form the hydroxylated intermediate IrO_2OH, resulting in the release of H^+ and e^-. This reaction indicates the early stages of surface oxidation, which may increase the likelihood of structural changes and oxidation beyond Ir_4^+, ultimately leading to the formation of higher oxidation states that promote dissolution. Eqn (13) depicts the dimerization and subsequent dissociation of the hydroxylated species IrO_2OH, resulting in $HIrO_2$ and O_2. Eqn (14) represents the further oxidation of IrO_2OH to IrO_3. The creation of IrO_3, particularly at high anodic potentials, indicates an oxidative environment that encourages Ir instability. According to Kasian et al. [94], IrO_3 formation is a critical factor in Ir dissolution because it introduces a higher oxidation state that is unstable in acidic media. As stated in equation (15), IrO_3 is further oxidized in the presence of water, resulting in IrO_4^{2-} and the release of protons. IrO_4^{2-} is a highly soluble species, which means that once Ir has reached this oxidation state, it can easily dissolve in the electrolyte. This dissolution significantly reduces the catalyst's active Ir content, reducing its OER activity. Furthermore, equation (16) shows the breakdown of IrO_3 back to IrO_2 with its expulsion of O_2. Although this mechanism can reduce Ir loss by renewing IrO_2, the repeated formation and decomposition cycles accelerate Ir degradation over time. In addition, the metallic Ir could be oxidized to Ir_3^+, resulting in catalyst degradation. Yu et al. [95] additionally found that soluble Ir ions can migrate through the Nafion 117 membrane, causing reprecipitation and affecting the anode/membrane interface and cathode. Recent work using ex situ ICP-MS analysis and detailed physicochemical characterization of the CL/membrane interfaces has advanced our understanding and accurate quantification of anode catalyst losses under realistic PEMWE operating conditions [96,

97]. Mechanical damage, gas bubble formation, and catalyst agglomeration all have an impact on CL stability. Limiting the size of the bubbles can significantly improve stability. While catalyst agglomeration, which involves particle growth mechanisms such as redeposition, Ostwald ripening, and coalescence, could result in electrochemically active surface area (ECSA) loss (Fig. 4c).

4.1.2. Metallic cations poisoning

Metallic cation poisoning from elements such as Fe_3^+, Cu_2^+, Al_3^+, and Ti_4^+ is a common issue in PEMWEs (Fig. 4d) [98]. The contaminants are commonly caused by feedwater, corrosion in water pipes and cell components, or other external sources. Upon entering the system, these cations infiltrate the CL, where they occupy ion exchange sites in both the electrolyte membrane and the ionomer. This causes increased overpotential at both the cathode and anode sites. Furthermore, these cations can block catalyst surfaces, particularly Pt at the cathode, resulting in metal or metal oxide precipitation [99]. However, this type of poisoning is usually reversible. The impurities are typically removed by immersing the MEA in a liquid acid solution, such as H_2SO_4 [90].

4.1.3. Catalyst support passivation

While investigating supported catalysts in PEMWE systems, the degradation mechanisms become more complex. One notable issue is passive action of the catalyst support, which is frequently caused by the formation of a poorly conductive oxide shell on the support surface. This shell contributes to increased ohmic resistance at the contact interface. Moriau and colleagues conducted a systematic investigation into the passivation of various supports commonly used to stabilize Ir for OER in acidic PEMWE [100]. Furthermore, it has been observed that certain impurities, such as fluoride ions (F^-), can dissolve TiO_2 supports during

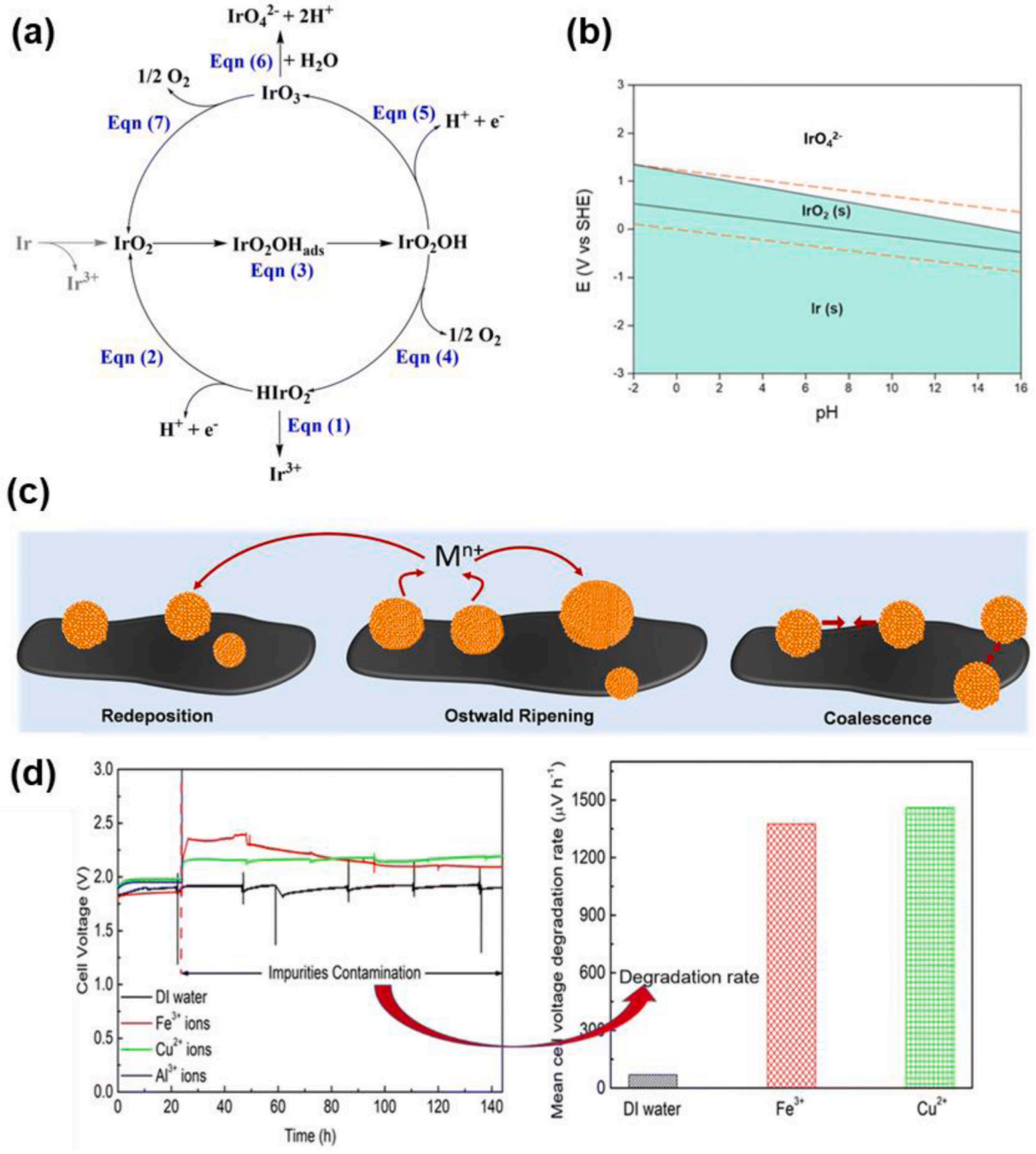


Fig: 4. (a) Proposed mechanism for Ir dissolution during OER adapted from Ref. [105] (b) Pourbaix diagram of Ir adapted from Ref. [93] (c) Proposed mechanism for catalytic particle enlargement adapted from Ref. [105] (d) Impact of cationic impurities on PEMWE performance adapted from [98].

polarization, complicating the degradation process [89].

4.2. Degradation mechanisms of membranes

The membrane in PEMWE systems serves two important functions: it separates the produced H_2 and O_2 gases and facilitates proton transfer between electrodes, completing the operational circuit. For optimal performance, the membrane in PEMWE must have outstanding chemical and mechanical stability, thermal resilience, high proton conductivity, and resistance to gas crossover [35,101]. As a result, the membrane is critical to the system's safety and energy efficiency. To ensure the optimal performance and longevity of PEMWE systems, a thorough understanding of the complex degradation mechanisms is required. Although numerous studies have been conducted on membrane degradation mechanisms in PEMFCs, research into PEMWEs is still in its early stages [102]. This section will investigate well-known degradation pathways, which primarily include chemical, mechanical, and thermal degradation.

4.2.1. Chemical degradation

Gas crossover, especially O_2 permeating through the membrane to the cathode, is an urgent problem in PEMWE systems. This process results in a two-electron reduction, yielding hydrogen peroxide (H_2O_2). When H_2O_2 interacts with contaminated metal ions like Fe_2^+, a Fenton reaction occurs, resulting in the formation of radical species (e.g., $HO^\bullet$, $HOO^\bullet$). These oxidizing agents aggressively attack the membrane's PFSA structure, causing significant degradation [103]. This deterioration shows as fluoride and sulphur release, membrane thinning, and enhanced mechanical failure, all of which exacerbate hydrogen crossover from cathode to anode, posing significant safety risks. Chandesris et al. created a 1D PEMWE model that includes membrane chemical degradation to study the effects of temperature and current density on this degradation. By analyzing the fluoride release rate in effluent water from both anode and cathode surfaces, they discovered that degradation occurs primarily on the cathode side and is heavily influenced by temperature. Current density has a less obvious impact: degradation rate is slow at very low current densities, increases to a peak at slightly higher current densities, and then decreases. Furthermore, metallic ions such as Fe_3^+, Cu_2^+, and Al_3^+ originating from corrosion and impurities in feed water can significantly impair the PFSA membrane [98,104]. These ions disrupt proton transport, resulting in higher ohmic resistance. For example, cell performance deteriorated during Fe_3^+ ion contamination tests, owing to Fe_3^+'s higher affinity for sulfonate groups in the Nafion membrane than H^+ [105]. This interaction reduces the effective triple-phase boundaries and proton conductivity.

4.2.2. Mechanical degradation

Mechanical degradation, caused by mechanical forces, frequently results in premature membrane failure. Long-term testing has shown that time-dependent deformation in Nafion membranes, including creep and microcrack formation, is a critical issue. This is caused by repeated stress on the membrane, which is exacerbated by uneven compression. Repeated membrane expansion and contraction weaken it over time, resulting in dimension changes, poor contact with electrodes, and pinhole defects. These pinholes, caused by localized stress, significantly reduce the life of PFSA membranes by causing cracks. Hydrogen crossover caused by PTL puncturing the membrane could result in intense local heating, gradually damaging the membrane. A polyethylene-based membrane that was designed as a cost-effective alternative to Nafion. Siracusano et al. [106] performed a long-term degradation test, beginning with a 200-h conditioning period at 1 A cm^{-2}, followed by 800 h at 3 A cm^{-2}, resulting in a mean voltage increase of 90 μV h^{-1}. The study concluded that continuous high current density operation, particularly above 2 A cm^{-2}, significantly contributes to membrane thinning. PEMWE is frequently used in industrial settings under high pressure, particularly in automotive applications that require storage pressures of several hundred bars. Mechanical failures, such as membrane creep, are more concerning than chemical degradation at these pressures, and gas cross-permeation becomes a serious problem. Finally, insufficient water circulation can cause premature system failure due to uneven water distribution or starvation. Excessive water flow can damage the system due to high pressure, while insufficient flow causes uneven distribution and hot spots. This causes irregular electric current and membrane swelling, which accelerates the degradation process.

4.2.3. Thermal degradation

The average working temperature for a PEMWE is 60–80 °C and membrane thinning is significantly accelerated at temperatures above 80 °C [107]. Thermal degradation is commonly associated with mechanical and chemical degradation. Temperature has a greater effect on membrane thinning than current density, according to extensive research. In this context, Chandesris et al. [108] investigated how temperature and current density affect membrane degradation. They analyzed the fluoride release rate in effluent water at both the anode and cathode during experiments conducted at temperatures of 60 °C and 80 °C at various current densities (ranging from 0.1 to 3 A cm^{-2}). The results showed a significant increase in membrane chemical degradation as temperatures rose. Current density has a smaller and more complex impact. At very low current densities, the degradation rate was very slow. However, as current density increased, the degradation rate peaked at relatively low current densities before decreasing. The observed reduction in degradation rate at high current densities is attributed to a lower molar percentage of O_2 on the cathode side. Guillet and colleagues have since conducted additional studies to back up their findings [109].

4.3. Degradation mechanisms of bipolar plates

The primary functions of bipolar plates (BPPs) in PEMWEs are to supply reactants (H_2O), discharge products (O_2, H_2), conduct electrons, and provide mechanical support. They must have low electrical resistance, provide insulation between the anode and cathode chambers to prevent gas mixing, be mechanically and chemically stable, facilitate fluid distribution, and have high thermal conductivity because they also help with heat transfer. Their multifunctionality, combined with the complex environment in which they operate, makes them highly susceptible to degradation and attenuation. At present, high-performance BPPs are typically based on titanium. They are still susceptible to hydrogen embrittlement, passivation, and corrosion, particularly at high current densities (43 A cm^{-2}) and strong oxidizing conditions [105].

4.3.1. Hydrogen embrittlement

There have been reports of hydrogen embrittlement in commercial pure titanium [110–112]. Some hydrogen permeation may occur as a result of the PEMWE's long operation. This permeating hydrogen can react with titanium-based electrodes to produce titanium hydride (TiH_2), a relatively brittle material. As hydrogen embrittlement progresses, the bipolar plates may crack due to high pre-tightening forces and operational pressures. Tawancy et al. [113] conducted a systematic study of hydrogen embrittlement in commercial purity Ti. They discovered that when exposed to gaseous hydrogen at elevated temperatures, grade 2 titanium readily formed hydrides and became extremely brittle, potentially leading to mechanical failure of BPPs.

4.3.2. Passivation

In high-potential, high-humidity, and O_2-rich environments, the thickness of the oxide layer on the surface of titanium BPPs gradually increases. This increase in oxide layer thickness causes an increase in contact resistance, which hinders efficient electron conduction [114]. This phenomenon has the potential to impact on the overall performance and (η) of PEMWE systems. As a result, titanium plates are typically coated with precious metals such as Pt or Au to meet the

durability and performance requirements of the high potential range and oxidative environment. Although the effective use of precious metals in PEMWE, their high cost, particularly for megawatt (MW) and gigawatt (GW) scale electrolyzers, severely limit their widespread practical application. As a result, there is an urgent need for research and development of low-cost coating materials to make PEMWE technology more economically viable and accessible [112].

4.3.3. Corrosion

Concerning anodes, titanium presently serves as the predominant material for electrode plates and gas diffusion layers (GDLs) due to its commendable corrosion resistance and electrical conductivity. However, in demanding high-potential and strongly acidic environments, titanium undergoes inevitable passivation. While the generated TiO_2 protective layer prevents further corrosion, its low electrical conductivity significantly elevates contact resistance. Moreover, F^- ions, resulting from membrane degradation, can disrupt the passivation layer, leading to subsequent corrosion. In the case of cathodes, titanium components exposed to saturated H_2 for extended periods are susceptible to hydrogen embrittlement [115]. This phenomenon involves hydrogen infiltrating the interstitial spaces of metal atoms, causing cracks and resulting in embrittlement and fractures. Coatings of noble metals (such as Pt, Au, and Ir) can effectively prevent corrosion of bipolar plates and reduce contact resistance, but their use increases costs. Therefore, a careful balance between performance and overall cost is essential [116]. When titanium BPPs are in a strong acidic environment and exposed to F^-, a reaction like the one described in Eqn (17) occurs, causing the corrosion resistant TiO_2 film on the surface of the BPPs to be eroded [111,117,118].

$$TiO_2 + 4H^+ + 6F^- \rightarrow TiF_6^{2-} + H_2O \quad (17)$$

5. PEMWE increasing efficiency

Significant endeavors have been invested in improving the effectiveness of PEMWEs.

5.1. Operation conditions

In a PEMWE, the major electrolysis process takes place in the stack, which involves gas contact and mass movement [119,120]. Several research, as shown in Table 3, have investigated the effects of various operating circumstances on PEMWE performance. These experiments focused on operational aspects such as input currents, pressure, temperature, flow rate, and density of current. Toghyani et al. investigated varied layout variables and working restraint, using a three-dimensional numerical model for evaluating PEMWE performance under steady-state settings [121]. They looked at temperature, pressure, thickness of the GDL, and the thickness of the membrane, as well as variations in current density, open-circuit voltage, and partial pressure of type that affect charge transfer rate. Choi et al. [122] studied the effect of cell polarity on PEMWE performance, emphasizing characteristics such as the quantity of channels. Pressure and temperature are critical operational variables for PEMWEs, affecting a variety of components [123,124]. Tjarks et al. investigated the connection between temperature and pressure in PtG systems with electrolyzer pressures of up to 20 bar, to enhance the PtG plant's overall energy demand [125]. Scheepers et al. studied the possibility of PEMWE membranes to enhance effectiveness [126]. In 2021, Afshari et al. developed a mathematical model for PEMWE that included electrochemical and core thermodynamic relations [127]. Their research focused on the overlap event, water transfer methods, and dispersion concentrations. The ultimate objective had been to maximize electrolyzer productivity while minimizing voltage loss. They investigated the effects of electrodes, BP, and membrane resistance on electrolyzer execution. The study also looked into the effects of membrane width, cathode tension, as well as heat on anodic hydrogen concentration. A thicker membrane reduced hydrogen crossing from cathode to anode, while increased cathode pressure increased crossover rate, with concentration and activation overpotentials leading 85 % and 3 %, respectively, at a current density of 10, 000 A/m^2.

5.2. Stack components

A cluster is made up primarily of the membrane, electrodes, catalyst, transport layers (TLs), and BPs. Fig. 5 depicts the schematic of a cell inside a PEMWE heap. However, as research and technology develop, specific elements may be added, modified, or removed from contemporary designs.

5.2.1. Membrane

The membrane is vital to the PEMWE cell stack, adding substantially to overall efficacy. The aforementioned component may have improved by using better-suited materials or applying innovative electrochemical and physicochemical approaches [128]. Using a coated membrane with catalysts such as iron (Fe) and nickel (Ni) on porous titanium [129] can efficiently decrease corrosion, increasing (η) and durability. Although PEMs are considered the weakest components in the PEMWE system, there are inadequate reports on PEM stability for PEMWE when compared to PEMFC. The harsher operating conditions in water electrolyzers compared to fuel cells affect PEM vulnerability over time. For example, a fully hydrated PEM can always stress itself through membrane swelling, resulting in decreased mechanical strength and integrity

Table 3
Operating conditions and their influence over the performance of PEMWE [137].

Main study	Operating conditions				Result/Suggestion
	Temperature	Pressure	Current	Transport	
Manufacturing	Yes	Yes			Elevated temperatures and pressures of around 30–45 bar should be the goals of every new activity
Bubble formation and porous structures				Yes	Synchrotron radiography observation of novel transport routes
Thermal and electrochemical performance	Yes	Yes	Yes		Optimal membrane size, GDL, and catalyst size for maximum sample performance
Water transport	Yes		Yes		Calculating the mass flow rate equation of discharged H_2O with H_2 in a cathode and the electro-osmotic drag coefficient
0-D simulation	Yes	Yes		Yes	Ohmic loss prediction while optimizing the conductive contact region
The pattern of anode channel and effect of gravity	Yes		Yes		Low flow rate and high operating temperature minimize ohmic and activation losses, and the catalyst's bubble coverage should be decreased
Heat and charge transport in the anode and membrane	Yes	Yes	Yes	Yes	Electrochemical performance criteria to achieve optimal accuracy
Anodic PTL			Yes	Yes	Measurements of the in-plane and through-plane permeabilities are necessary to accurately describe PTL structures

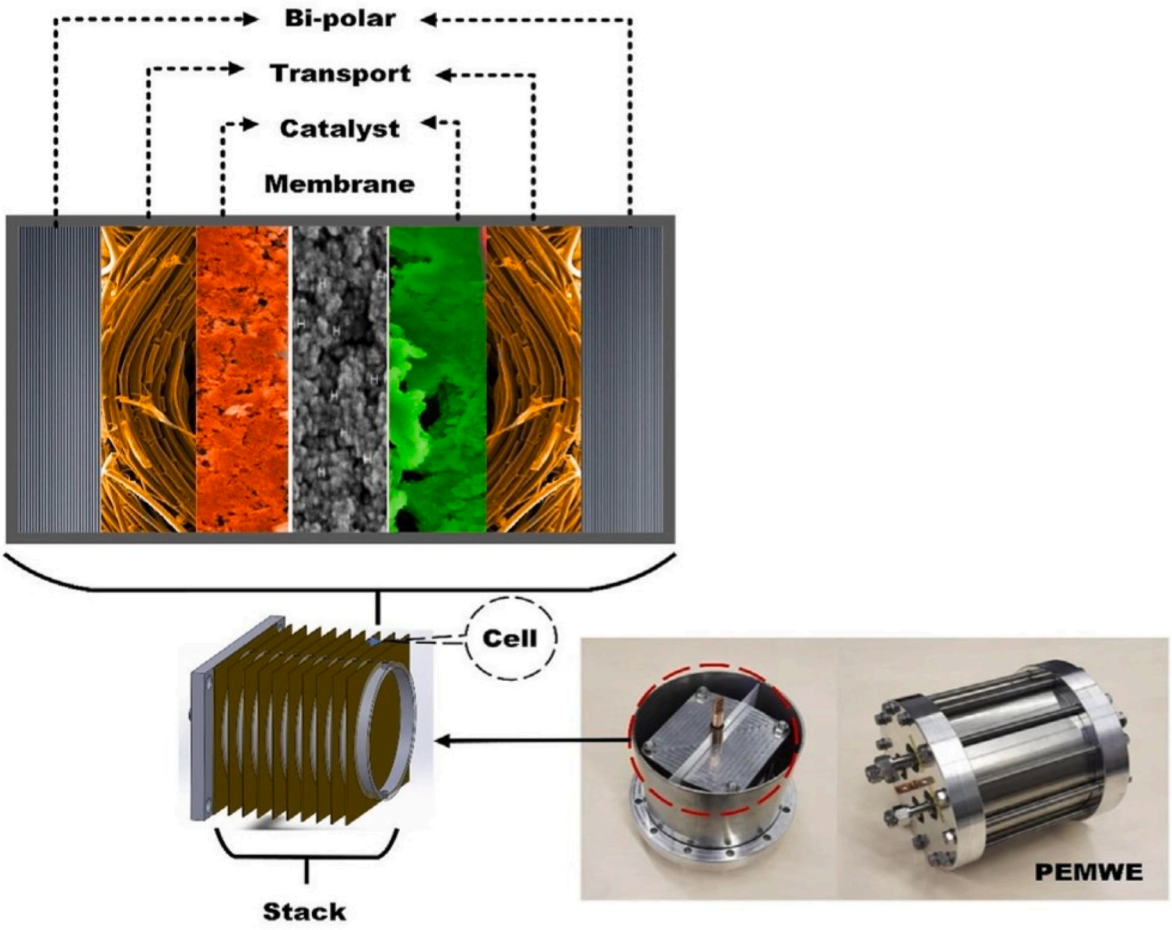


Fig: 5. Illustration of PEMWE stack components adapted from [137].

of the PEMs.

The ideal proton membrane should have beneficial characteristics like high conductivity, durability, and chemical stability. Common proton membranes consist of polystyrene sulfonic acid, poly(trifluorobenzene) sulfonic acid, phenolic resin sulfonic acid, and perfluorosulfonic acid (PFSA). Typical commercial proton membranes include Nafion, Flemion, Aciplex-F, 3 M, and SCC [23,101]. However, such commercial PEMs typically fail to deliver satisfactory results under harsh operating conditions such as high temperature (>100 °C) and low humidity (<50 % relative humidity), necessitating the development of novel polymer membranes. Small molecules such as heteropolyacid, silica, and titania are commonly used in this process. Zhang et al. [130] Used phosphotungstic acid (PWA) doping in polyethersulfone-polyvinylpyrrolidone (PES-PVP) to create a high-temperature resistant composite film with enhanced proton conductivity and mechanical strength. Barati and colleagues added TiO_2 nanoparticles to a composite membrane made with polybenzimidazole (PBI) and lignin as base materials, which significantly improved the original composite membrane's high-temperature acid retention and conductivity [131]. Furthermore, introducing a second polymer is an effective method for producing high-performance proton membranes. Siracusano et al. [132] Investigated PFSA aquivions short side chain reinforced membranes with an equivalent weight of 980 g eq.$^{-1}$ and a thickness of 50 mm produced by Solvay specialty polymers and introduced expanded PTFE fibers to reinforce the membranes. Following 3500 h of durability testing, the polarization resistance was reduced from 760 mΩ cm^2 to 320 mΩ cm^2, and the active surface area loss was reduced from 48 % to 35 %, significantly improving mechanical stability and proton conductivity. The improved performance can be attributed to more critical expanded PTFE fibers formed via ionic dispersions. The obtained membranes demonstrated excellent stability and durability, with performance remaining largely unchanged after 3500 h of continuous testing.

The composite membrane demonstrated reduced membrane thickness while improving mechanical and dimensional stability. Composite membranes can also be created by presenting small molecule monomers that undergo cross-linking reactions during the fabrication process. This means that the formed polymer can retain much of its original morphology and structural integrity in the composite membrane network, resulting in improved membrane efficiency. Liu and colleagues developed a new double cross-linked composite membrane using polyionic liquids (PILs) and polybenzimidazole (NbPBI) as substrates, with the addition of trimethoxysilane groups ([TSPDO]BrCl) effectively mitigating ionic liquid leaching and SiO_2 agglomeration in the membrane [133]. As a result, the membranes prepared using this method have high proton conductivity and mechanical stability, which is expected to improve PEMWE applications at high temperatures. Huang et al. [134] used an ultra-high molecular weight polyethylene (UHMWPE) with high processability and modulus to create a thin porous membrane of ≈300 nm via biaxial stretching and then impregnated the membrane with Nafion solution to produce an ultrathin PEM. The resulting PEM had a significantly thinner thickness than commercial Nafion membranes (1.6 vs. 25.4 μm), but improved mechanical and electrochemical properties, indicating a new direction for subsequent studies.

In conclusion, PFSA-type plasmonic membranes are the most widely used commercially, and they already provide the majority of the benefits. The development of plasmonic membranes that can withstand harsh

environments such as high temperatures and low humidity is currently the focus of research. In addition to incorporating small molecules into existing feedstocks, it may be possible to improve plasmonic membrane performance by introducing additional polymers [135]. However, the polymeric proton membranes studied thus far have the disadvantage of being more expensive, and other polymers must be explored to reduce the cost of PEMWE. It is additionally essential to consider other electrochemical or physicochemical techniques when developing novel proton membranes. Secondly, loading membranes with iron, nickel, and titanium to reduce membrane corrosion can aid in the commercialization of PEMWE [136].

5.2.2. Catalyst

The catalyst layer (CL) of catalyst-coated membranes (CCMs) is a further significant element in PEMWE cells. Presently, nearly all of PEM electrolyzers use costly PGM catalysts, which are recognized for their high price and susceptibility to decline [95]. As a result, there is a rational tilt to substitute these expensive and valuable catalysts with cheaper [138], frequently accessible resources from the Earth, including molecular catalysts [139], metal cobalt phosphide (CoP) [140], MoS_2--based materials, and $[Mo_3S_{13}]$ clusters placed to N-doped carbon nanotubes [141]. Developments in technology also result in significant advances in compounds such as Iridium, Titanium, and Platinum, including unique versions. Examples of such advances include iridium core/shell catalysts via galvanic exchange, nano-size IrOx catalysts with excellent activity and stability [142], $NeTiO_2$ nanofibers [143], and the use of aerogel supports. Individual components do not alone determine cell performance and endurance; fabrication procedures also have an impact. Furthermore, factors during manufacturing have a considerable impact on CL shape. Changes in ionomer amount or dispersion have a significant effect on kinetic performance, according to the findings. Greater ionomer concentrations and poor ionomer uptake (catalyst-ionomer segregation) hindered kinetics, most likely due to the ionomer having limited access to the Ir sites. Differences in catalyst layer thickness also had an adverse effect on ohmic loss, presumably by decreasing the catalyst layer-PTL interface and raising contact resistance. To enhance CL engagement and improve catalyst service, particularly at high current density, transport resistance must be reduced while protonic and electronic conductivities are increased [144]. This principle also includes electrolytes, and regulating the present amount improves conductivity and general performance [145].

As a result, enormous efforts are being made to develop Pt-free electrocatalysts that can overcome the disadvantages of Pt-based catalysts. Muntean R. et al. [146] Designed a graphitic carbon nanofiber-supported ternary alloy catalyst (PtCoMn/CNF) for HER in acidic media. The developed PtCoMn/CNF electrode has a lower overpotential of 250 mV at 40 mA cm^{-2} and a Tafel slope of 120 mV dec^{-1} in 0.5 M H_2SO_4, comparable to platinum-based electrodes (Pt/CNF). Holzapfel P. K. et al. [141] created a noble metal-free, efficient, and long-lasting electrocatalyst of $[Mo_3S_{13}]^{2-}$ clusters supported on nitrogen-doped carbon nanotubes (Mo_3S_{13}-NCNT) for the HER and tested its electrochemical performance in a single-cell PEM water electrolyzer. The developed Mo_3S_{13}-NCNT demonstrated an elevated current density of 4 A cm^{-2} at a cell voltage of 2.36 V, with reasonable stability (83 mV h^{-1}) during a 100-h durability test. Lazer et al. [147] described reduced graphene oxide-supported Au and PdAu nanocomposites (PdAu/rGO) for the HER in acidic media. The PdAu/rGO electrocatalyst demonstrated high electrocatalytic activity and stability, with an overpotential close to zero and a Tafel slope of 46 mV dec^{-2} despite a low Au and Pd atomic percentage.

Tajuddin A. A. H. et al. [148] also reported on precious metal-free graphene-encapsulated NiMo composites as an acid-durable HER electrocatalyst. In single-cell studies, the prepared graphene-encapsulated NiMo composite demonstrated exceptionally stable electrocatalytic activity over 10,000 cycles and 100 h of durability at a fixed cell voltage of 2.2 V. Xie Z. et al. [149] investigated defect-rich 1T-2H MoS_2 nanosheets (NSs) as HER electrodes with an extremely low load of 0.14 mg cm^{-2}. The results showed an extremely lower overpotential of 192 mV and a smaller Tafel slope of 44 mV/dec^{-1}. In single-cell PEM water electrolysis experiments using the Nafion-115 membrane, a cell voltage of 2.25 V was observed at 2 A cm^{-2}. Dharman R.K. et al. [150] used a solvothermal technique to create nickel-stimulated mixed-phase (1T/2H) MoS_2 and investigate it's HER performance in an acidic medium. The developed nickel (9 %) stimulated MoS_2 showed superior electrocatalytic activity in a 0.5 M H_2SO_4 electrolyte. It achieved an overpotential of 127 mV to reach a current density of 10 mA cm^{-2} and exhibited a smaller Tafel slope of 86 mV dec-^{1}. P'erez G. et al. [151] developed a carbon black-decorated Rh-based organometallic catalyst for the HER with an extremely low metal loading of 0.0066 mg cm^{-2}. Rh/C's electrocatalytic activity was evaluated using the RDE method in an acidic environment (pH^{-1}) and then tested in a PEM water electrolyzer. The results demonstrated a competitive cell voltage of 1.9 V with 0.5 A cm^{-2} at an operating temperature of 70 °C, with an ultra-low metal loading of 0.0066 mg cm^{-2} on MEA. Furthermore, these results are found to be equivalent to those obtained with conventional Pt-based catalysts under comparable operating conditions. Huang et al. [152] presented the development of Ru nanoclusters anchored on N-doped carbon with abundant micropores, which demonstrated remarkable activity for the HER across the pH spectrum. The Ru NCs/NC catalyst had notably low overpotentials, measuring 14 and 30 mV at a current density of 10 mA/cm^2 in alkaline and neutral conditions, respectively, while demonstrating performance comparable to Pt/C in acidic environments.

Commercial PEMWEs have high Ir loading (>2 mg cm^{-2}). Although low Ir loading is traditionally regarded as a difficult mission due to significant performance/durability reductions, Ir loading can be reduced by using a new catalyst structure [17]. To address the challenges of iridium scarcity and high costs, academic and research organizations are focusing more on reducing iridium loadings and investigating potential replacements for iridium-based catalysts. Hao S. et al. [153] developed a highly efficient OER catalyst of $W_{0.2}Er_{0.1}Ru_{0.7}O_{2-\delta}$ for PEM water electrolysis. This electrocatalyst has a low overpotential of 168 mV at 10 mA cm^{-2}, a high durability of 500 h, and a small Tafel slope of 66.8 mV dec^{-1} in acidic media. Furthermore, it operated steadily for 120 h at a constant current density of 100 mA cm^{-2} in a single-cell PEM water electrolyzer. Min X. et al. [154] developed a highly efficient IrOx/W–TiO_2 electrode material for the OER in PEMWE to lower the precious metal content. In both half-cell and single-cell studies, the IrOx/W–TiO_2 electrode demonstrated higher mass activity and electrocatalytic activity than commercial IrO_2. It was achieved with a cell voltage of 1.602 V at a current density of 1 A cm^{-2} while reducing precious metal loadings to # 0.2 mg cm^{-2}. Zhang J. et al. [155] presented Ru@Ir–O core-shell nanostructures with integrated O_2 and tensile strains in the Ir shell. At a current density of 10 mA cm^{-2}, these structures had a low overpotential of 238 mV for OER under acidic conditions. The Tafel slope was reduced to 92.6 mVdec^{-1}. Furthermore, the proposed electrocatalyst demonstrated 78-fold higher mass activity than conventional IrO_2.

Raja D. S. et al. [156] created a noble metal-free electrocatalyst of $FeCo_3O_4$@C/FTO for the OER and tested its performance in 0.5 M H_2SO_4. The catalyst demonstrated remarkable electrocatalytic activity in 0.5 M H_2SO_4 with a lower overpotential of 396 mV at 10 mA cm^{-2} and a small Tafel slope of 68.6 mV dec^{-1}. It also demonstrated stable electrocatalytic activity during 50 h of continuous chrono-potentiometric operation. Joo J. et al. [157] recently developed a highly efficient and long-lasting electrocatalyst for PEM water electrolysis using Mn–RuIr NCT heterostructures. The developed Mn-RuIrNCT exhibits exceptional OER activity and stability, with an ultralow overpotential of 198 mV at 10 mA cm^{-2} and 247 mV at 100 mA cm^{-2}, as well as a smaller Tafel slope of 45 mV dec^{-1}. This is significantly smaller than that of the control catalysts (IrO_2, RuO_2, and RuIr NPs). In general, a smaller Tafel slope and lower overpotential indicate that the OER has faster kinetics.

The chronopotentiometry method was used to study durability at a constant current density of 10 mA cm^{-2} for 180 h in 0.1 M $HClO_4$ electrolyte. The results showed that the device remained stable throughout the 180-h period. The Mn–RuIr NCT catalyst demonstrated improved electrochemical performance in a single-cell PEM water electrolyzer, with a current density of 1 A cm^{-2} at 1.698V compared to pure IrO_2 (1.775 V). In addition, extensive research has been conducted on Ru-, Ir-, and noble metal (NM)-free electrocatalysts for the OER under acidic conditions.

Pham et al. [158] synthesized IrO_2-coated TiO_2 core-shell microparticles (IrO_2@TiO_2) using a simple and scalable method. At 1.67 V, the catalyst achieved current densities of 1 A/cm^2 with loading as low as 0.4 mgIr/cm^2. Furthermore, extensive research is being conducted to develop more cost-effective catalysts, such as Ru-based OER catalysts, because of their low cost, widespread availability, and superior intrinsic activity [159,160]. However, its practical application is limited by poor dissolution resistance under acidic conditions, resulting in a short lifespan of only a few hours. To address this, researchers have focused on two strategies: (1) incorporating electron-donating elements to reduce Ru's oxidation state, and (2) increasing the catalyst's intrinsic activity to lower its operating potential, thereby extending its operational life [161]. Wang et al. [162] improve the activity and stability of Ru-based catalysts for PEMWE by tailoring the reaction pathway with MRuOx (M = Ce, Sn, Cr, etc.) solid solutions. Surprisingly, the SnRuOx catalyst has exceptional durability and catalytic efficiency, achieving a low voltage degradation rate of 53 μV h^{-1} over 1300 h at 1 A/cm^2, outperforming both conventional ruthenium-based and certain iridium-based catalysts.

5.2.3. Gas diffusion layers

The gas diffusion layer (GDL) is the subsequent component of the PEMWE. There is a long history of study on GDL, indicating significant potential for progress and cost-cutting efforts [163]. The formation of O_2 gas in PEMWE's transition layers is a substantial challenge to increase efficiency. To ease mass transfer and improve electrical conductivity, the PEMWE design has a porous layer called the PTL. Ongoing research seeks to optimize the benefits of PTL [164]. Lee et al. collected investigations on PTL mass transport limitations in PEMWE to improve (η). PTL improvements can reduce mass transport losses, increase catalyst usage, and lower ohmic and kinetic losses in PEMWE [165]. A mechanism known as mass transport limitation (MTL) is a prevalent issue in the circulation of the PEMWE cell potential at high current density, resulting in lower (η) due to impeded transport in the PTL. Panchenko et al. [166] investigated this phenomenon extensively utilizing neutron-based imaging techniques, discovering various in-situ stoichiometries during the absorption of polarization curves. The significance of their research is in maximizing effective cell design. Kim et al. [167] review till 2022 provides useful insights on PTL and CL techniques and materials, including a novel way to merge PTL and CL via bilayer titanium mesh PTL in a PEMWE.

Seweryn et al. [168] emphasized the functional relationship between liquid water content and electrode distance in the PTL, and an unforeseen water content mode was observed in anodic porous materials, revealing the reason for the mass transfer limitation in the PEMWE. Bromberger et al. [169] investigated the pore size distribution of the PTL employing capillary flow porosimetry and discovered that capillary pressure has a significant effect on mass transfer limitation in PEM electrolysis cells with a flow field. The gas transportability increased as the gas flow rates increased, as did the current density. Pushkarev et al. [170] investigated the effects of PTL parameters on electrolytic cell performance and found that using a less porous PTL increased mass transfer loss while the excellent porous electrode design facilitated high current density. Zhao et al. [171] developed a super hydrophilic PTL that significantly reduced mass transfer overpotential due to decreased gas saturation in the anode PTL, resulting in increased electrolyzer efficiency. Parra-Restrepo et al. [172] investigated the effects of average pore sizes and porosities on electrolytic cell properties. The best properties were achieved with a mean pore size of 10 μm and a porosity of 0.31. Gas-water transfer was limited when using a PTL with an average pore size of 3 μm.

Wrubel et al. [173] created a mathematical model of the PEMEC using a novel PTL that combines electrochemistry and multiphase transport. They discovered that larger pore sizes improved water access to the anode catalyst layer (ACL) as well as O_2 gas transfer. Chen et al. [174] created a multi-physical PEM electrolysis cell model to investigate the liquid water distribution in the PTL, local current density, and cell properties. The numerical analysis revealed a significant water shortage under land for the 100-μm-thick PTL with a current density of 5 A cm^{-2}. The water shortage caused the voltage to rise by 0.2 V, resulting in a significant change in local current density. To summarize, fine-tuning the microstructure of the PTL (pore size and porosity) can affect the mass transport and contact resistance of the PTL/CL, thereby influencing the electrochemical performance of PEMWE. In broad terms, high current density produces a large amount of O_2, which prevents water from accessing the catalytic site in a timely manner, so two-phase (gas/water) transport reduces porosity and, as a result, the transport resistance of the PTL [175]. The use of porous electrodes with large surface areas is an effective method for improving mass transport, and the addition of PTL can further improve the mass transfer limitation in the PEMEC.

5.2.4. Bipolar plate

BP is the final critical component of PEMWE. This adaptable device distributes air and liquid uniformly, controls the electrical current between cells, adjusts temperature, and prevents gas and coolant leaks [176]. To sustain the stack's operational conditions, BPs must have low interfacial resistance to contact and high corrosion resistance. Furthermore, a layer of coating may be required for BPs [177]. BPs are a fairly costly component due to their comprehensive requirements [178]. However, given the current emphasis on lowering expenses for stack components in PEMWEs, there is an increasing interest in investigating inexpensive components coated with anti-corrosive layers as viable substitutes for traditional BPs. This technique seeks to reduce expenses while, ideally, improving performance [179–181]. Taner et al. [182] assessed a prototype HyPro system employing PEMWE and found that it could create H_2 approximately 4.5 times with greater efficacy than other systems. PEMWE with a magnet (Cr–C coated SS304 BP) has a greater (η) compared to the bare one. Electrolyzers, BPs, and certain integrated elements are now manufactured thanks to advances in technology for production and 3D printing [183].

In order to minimize the cost of bipolar plates, we must overcome the technical and cost challenges of PGM-coated Ti by exchanging the coating and core materials with replacements that provide excellent corrosion/chemical resistance, good through-plane electrical conductivity, and low interfacial contact resistance (ICR). Instead of Au and Pt, layered structures made of metal and metal nitride (TiN, TaN, ZrN, CrN, and NbN) or metal carbide (TiN–C) have been extensively studied [184–186]. Ye et al. [184] evaluated the ICR of three metal nitride coatings on Ti metal and found that the Ta/TaN coating (3.9 X cm^2) obtained the least ICR value, a 74 % decrease over pristine Ti metal (14.9 X cm^2). At this point, Laedre et al. [187] investigated different non-coated metal plates used as bipolar plates in PEMWE in terms of current density, corrosion reaction, oxide formation, and ICR change during stability tests. Ta, Nb, and Ti demonstrated high corrosion resistance with low current density and mass change; however, Ta and Nb showed a substantial rise in ICR because of the development of a surface oxide layer over extended CA tests at 2.0 V.

Stainless steel samples (304 L and 316 L) and tungsten (W) showed minor corrosion caused by OER activity, however, required to be modified to meet the demand of a bipolar plate for PEMWE. Therefore, stainless steel is considered an alternative to Ti metal in PEMWE due to its ample supply and ease of machining; however, extremely low pH can cause the dissolution of stainless steel, leading to a decline in MEA effectiveness during long-term operation [188]. Fiedler et al. found that

stainless steel 316 L degrades significantly, with a high dissolution rate measured by a scanning flow cell setup coupled to an inductively coupled plasma spectrometer, but contamination in a full cell is estimated to be less than 1 ppm [189]. Becker et al. [118] also recommended using carbon-coated stainless steel as a bipolar plate instead of Ti, citing no proof of deterioration in stainless steel after 30 days of testing at 2 A/cm^2. The findings indicate that stainless steel can be used as a core material for bipolar plates in PEMWE if a covering material is used to retard continuous dissolution caused by the absence of a passivation layer on stainless steel.

Rojas et al. [181] used physical vapor deposition to prepare various surface-coated (CrN/TiN, Ti/TiN, Ti, and TiN) stainless steels and evaluated their corrosion behavior to determine the best coating material for stainless steel. While investigating corrosion behavior in terms of weight loss, dissolved metal amount, and surface change, they discovered that the Ti/TiN layer has the best stability and ICR value after the corrosion test (9.9 mX cm^2). Not only TiN, but also metal nitride, metal alloys, and even conductive polymers have been studied to enable the use of stainless steel as a bipolar plate in PEMWE [190,191]. Unexpectedly, Stiber et al. [192] tested the long-term stability of Nb-coated stainless-steel bipolar plates for 14000 h, which functioned with only a 5.5 μV h^{-1} average degradation rate (operating at 1.9 V while maintaining 1 A/cm^2). The addition of Nb produced weakly conductive amorphous Nb oxide, it had no effect on the performance of the PEMWE cell, implying that Ti could be completely replaced with covered stainless-steel bipolar plates. Cu is also used as a substitute for Ti metal, but it requires a 30 μm thick Nb metal layer to prevent Cu corrosion [193].

5.2.5. Charge transfer in cell components

The charge transfer coefficient (CTC) is critical for forecasting the current-voltage characteristics of electrolyzers and provides useful insights into electrode parameters. Biaku et al. investigated the temperature dependency of the O_2 electrode CTC in an industry-standard PEMWE stack. Tijani et al. investigated the effect of operating temperature on CTC along with its consequences for PEMWE operating voltage, which is especially important in PtG systems because activation overpotential determines the operating voltage. Their results revealed that CTC values improved at greater operating temperatures, with anode electrodes experiencing more of an impact than cathode electrodes. Surprisingly compression had no substantial effect on CTC at each electrode [194]. Then they investigated the connection among exchange current density, CTC, and PEMWE performance. Their findings revealed that lower CTC demands are connected to greater exchange current density, which leads to less activation overvoltage and, as a result, reduced operating voltage systems. In essence, using overvoltage in PEMWE has shown to be more efficient, allowing for the best use of extra energy capacity in this design. They also investigated the temperature-CTC interaction, focusing on the O_2 electrode [115]. The compression and high temperatures can change the reaction environment in complicated ways. The Arrhenius equation provides the evidence that rate of reaction in an electrochemical process depends on the temperature and pressure. The CTC may change based on how temperature alters the symmetry of the energy barrier in the electrochemical reaction, and as the temperature rises, the activation energy barrier becomes easier to overcome, resulting in faster reaction kinetics. On the other hand, compressing the electrode materials (porous material), increases the surface area in contact with the electrolyte, potentially improving the charge transfer coefficient and reaction sites. However, over-compression can result in reduced porosity or clogging, which restricts reactant mass movement and slows down the pace of reaction. Therefore, to maximize the charge transfer coefficient for durability and efficiency in water electrolysis process, a need to balance the compression and temperature of reaction. A deeper understanding of these relationships could lead to the development of more efficient electrolyzers, minimizing energy losses during operation. As research progresses, it will be essential to continue investigating how different parameters interact with temperature and CTC and will inform the design of next generation electrolyzers aimed at improving performance and durability in various applications.

6. Market of electrolysis

Hydrogen is undergoing a renaissance, major financial institutes are positioning themselves to advise on H_2 in anticipation of a growing commercial market [116,195,196]. The European Union's 2020H_2 strategy demonstrated an intense dedication to innovation, aiming to deploy 40 GW of electrolyzers over the next generation [197]. In response, the industry is expanding its manufacturing capacity, establishing 'giga-factories' similar to large-scale lithium-ion battery production plants [198,199]. To implement these ambitions and integrate H_2 as an accepted component of the worldwide energy system, hydrogen must be cost-competitive with existing technologies.

6.1. Commercial status of electrolysis

Currently, electrolysis accounts for just around 1–2 % of worldwide hydrogen generation or approximately 7 million tons per year [200]. However, this share is expected to grow in the future. Fig. 6a depicts the global installed hydrogen electrolyzer capacity, which has grown significantly over time, with near-term estimates indicating an upward trend from diverse origin [201–204]. Over the last decade, worldwide production has risen swiftly, with a median annual growth rate of 32 % since 2010.

AWE became the dominant innovation, accounting for more than 90 % of global capacity in 2010. Nevertheless, its annual growth rate has slowed down to 19 % since then. In contrast, PEMWE capacity has increased at an astonishing 80 % per year, surpassing AWE's installed capacity in 2019 alone [26].

6.2. Markets for hydrogen

Historical excitement about the promise of hydrogen is not a new trend. Hydrogen technologies have continuously followed the Gartner-Hype Cycle model [205], which is defined by cycles of high expectations, eventual disillusionment, and occasional bankruptcy. The prospective hydrogen market is shifting due to rising rivalry with other low-carbon methods. However, their future is currently bleak, with charged-powered automobiles [206] and electric heat pumps gaining traction in the transition away from fossil fuels. Fig. 6(b and c) depicts analysts' forecasts for the competitive stance of hydrogen in different market's income [204,207].

6.3. Major manufacturers of electrolyzers

The global electrolyzer market has a somewhat consolidated structure. According to the recent study in 2023 found just 26 small to big-scale suppliers, with 13 in AWE, 11 in PEMWE, and 2 in SOEC (Table 4) [208]. However, this picture is vulnerable to alter due to the market's dynamic character, as demonstrated by Cummins' acquisitions of Hydrogenics and Air Liquide [209]. Entities such as IRENA [210], the ETC [204], and numerous commerce study companies extensively address significant equipment producers.

7. Current capital cost of electrolyzers

The financial sustainability of hydrogen electrolysis, unlike many other sectors of the energy sector, is heavily dependent on CAPEX. The two primary factors influencing hydrogen price were the investment cost of the electrolyzer and the operating expense of electricity as the input fuel, which varies significantly depending on location, technological advances, and other temporary factors. This section analyzes the

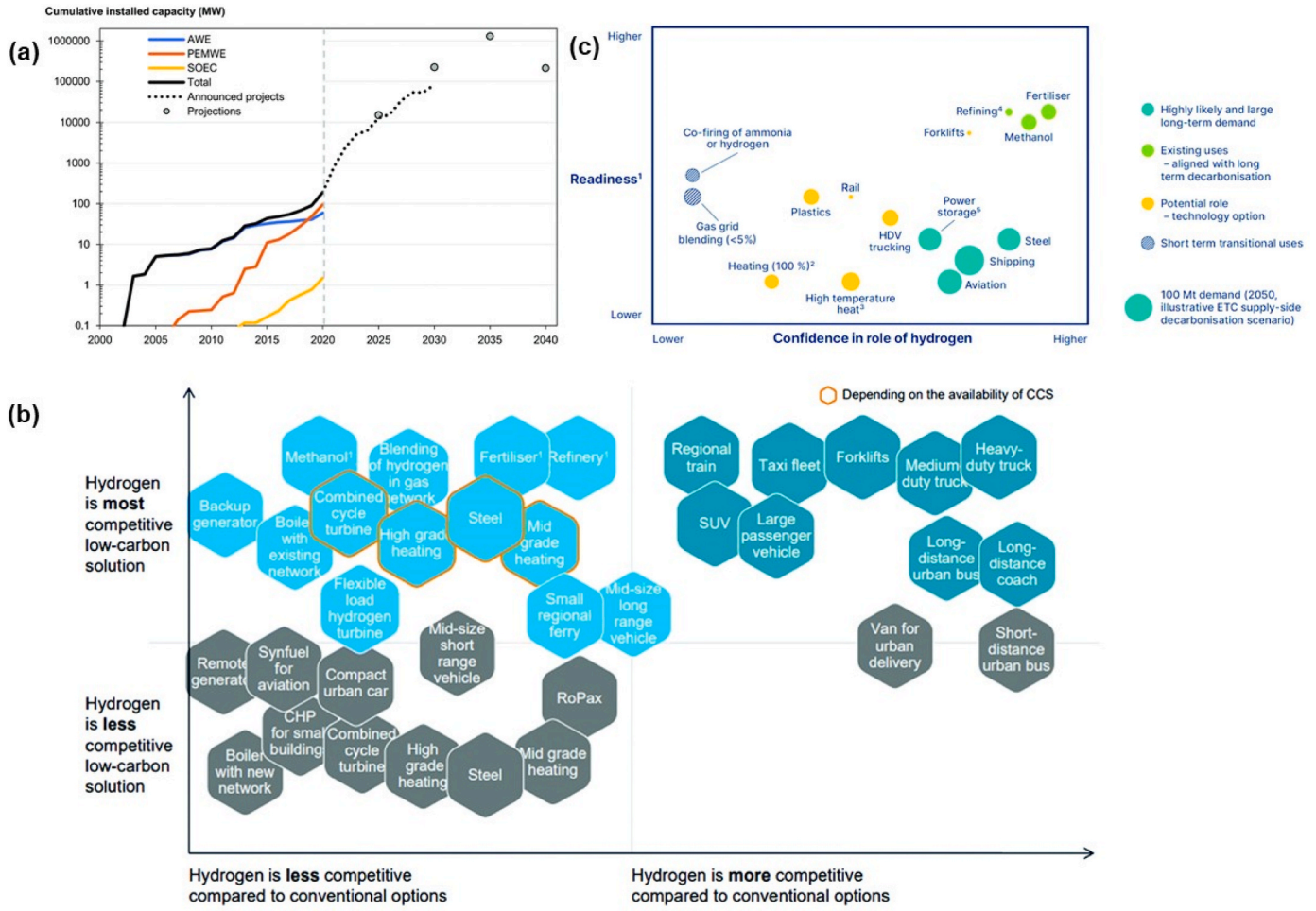


Fig: 6. (a) The total installed capacity of contemporary hydrogen electrolyzers, split down by techniques, along with market size predictions made by analysts adapted from Refs. [26,201–204] (b) Applications of hydrogen are competitive when compared to traditional and low-carbon alternatives adapted from Refs. [26, 207] (c) The evaluation of several possible applications of hydrogen carried out for the ETC by SYSTEMIQ adapted from [26,204].

hydrogen industries and the industry's anticipated expansion, with a special focus on the present capital costs of electrolysis devices and their reliance on equipment and production phases. The investigation also includes expected capital cost trends over time, as well as an examination of the elements that may contribute to cost savings. The cost of electrolyzers is critical to their performance and competitiveness when compared to other techniques of hydrogen and low-carbon fuel creation. Assessing the price of electrolyzers poses issues for various reasons: (1) the methodology is still in its infancy of mainstream development purposes, leading to an absence of easily accessible data; (2) costs vary significantly based on technology, influenced by design, material requirements, and the maturity and scale of production; (3) prices exhibit substantial variations based on the country of production, especially between China and other areas; (4) costs are changing, going through rapid shifts as producers expand up their product.

7.1. Survey of current electrolyzer costs

Current electrolyzer cost estimates range from $170 to $2300 per kilowatt (kW) of operation capability (Fig. 7). Reported expenses for AWE range from $170 to $1000 kW^{-1}, PEMWE from $700 to $2000 kW^{-1}, and SOEC at around $2000 kW^{-1}. Multiple organizations have pointed out the minimum price for alkaline electrolyzers at $170 kW^{-1} ($200 kW^{-1}), particularly concerning claimed prices from current Chinese manufacturing facilities. Fig. 7a demonstrates an obvious trend of significantly slipping expenses in recent years. Bloomberg New Energy Finance (BNEF) predicts that the CAPEX for large-scale electrolyzers experienced a reduction of 40–50 % in the five years leading up to 2019 [211]. Notably, AWE costs dropped from $2000 to $1200 kW^{-1} during this timeframe, while PEMWE costs decreased from $2800 to $1400 kW^{-1}.

7.2. Influence of materials and components

AWE and PEMWE techniques are considered to be fairly mature, with products that are marketable and known prices. Nevertheless, SOEC added increased cost fluctuation and uncertainty, surpassing 1 MW of installed capacity only in 2019. Costs for novel methods such as AEMWE and PCCELs remain uncertain due to a lack of enormous-scale comparison systems. Electrolysis systems include more than only the electrolyzer stacking. Auxiliary components, called the balance-of-plant (BoP), comprise the conditioning of power (transformers and converter for DC supplies), treatment of water (cleansing and warming), and hydrogen purifying (separation, drying, and pressurization). These parts are mature technologies that are widely used across an array of industries. The electrolyzer stack's contribution to total system cost ranges between 27 % and 64 %, according to the literature. Fig. 7d shows multiple projections of the capex contribution resulting from various electrolyzer components. For example, IRENA estimates that the stack accounts for 45 % of the entire system cost, with the remainder divided between the electricity supply (28 %), water circulation rate (12 %), hydrogen processing (11 %), and the cooling (4 %) [210].

Table 4
Technical specifications of commercially available electrolyzer products adapted from [208].

Supplier	Technology	Model	Capacity (Nm^3/h)	Pressure (bar)	Consumption (kWh/Nm^3)	Ramp up (s)
Asahi Kasei	AWE	–	up to 2000		–	–
Angstrom Advanced	AWE	–	up to 2000	16	4.2	–
Auyan	AWE	AOQING	up to 1000	16	4.4	–
Bloom Energy	SOE	–	up to 500	1	–	–
Cummins	PEM	HyLYZER	up to 4000	30	4.3	–
Elogen	PEM	–	up to 2000	30	4.8	–
Green Hydrogen	AWE	HyProvide X-series	up to 1200	35	4.8	–
HydrogenPro	AWE	–	up to 1000	30	4.4	–
Hystar	PEM	Miral200	up to 1200	4	4.4	30
H-TEC systems	PEM	MHP	up to 2131	15–30	4.8	30
ITM	PEM	Poseidon	up to 4000	30	5.5	–
John Cockerill	AWE	DQ1000	up to 1000	30	4.3	–
Longi	AWE	Lhy-A1500	up to 1500	16	4.4	–
McPhy	AWE	McLyzer 3200-30	up to 3200	30	4.7	30
Nel	AWE	A3830	up to 3880	1	4.4	–
Nel	PEM	M5000	up to 4920	30	4.5	–
Next Hydrogen	AWE	NH-500	up to 500	10	5.6	20
Ohmium	PEM	Lotus	up to 700	34	–	–
Peric	AWE	CDQ	up to 1500	15–20	4.3	–
Plug Power	PEM	EX-4250D	up to 2000	40	4.5	<60
Proton Onsite	PEM	M400	up to 400	30	5	–
Siemens Energy	PEM	Silyzer 300	up to 3700	30	5.1	<60
Sunfire	AWE	HyLink	up to 2165	30	4.7	–
Sungrow	PEM	SHT500P	up to 500	30	–	–
Thyssenkrupp	AWE	–	up to 4000	1.3	4.4	–
Topsoe	SOE	–	up to 32000	2	3.1	–

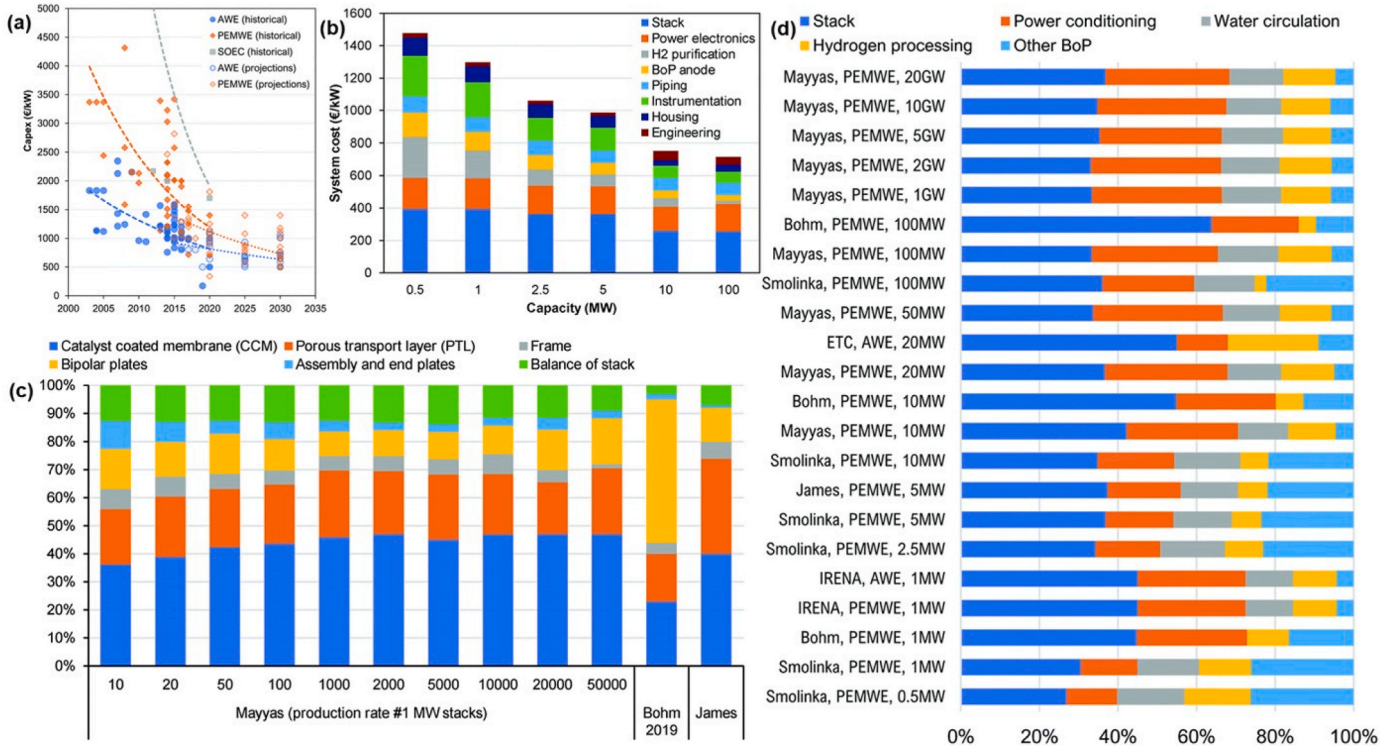


Fig: 7. (a) Electrolyzer's CAPEX cost adapted from Refs. [26,214–216] (b) Component contribution to the cost of the PEMWE electrolysis system at varying capacities adapted from Refs. [26,214] (c) Cost contribution estimates for various PEM electrolyzer stack elements adapted from Refs. [26,212,217,218] and (d) Cost comparison of various electrolyzer's components adapted from [26,204,212,214,217].

As production scale increases, the balance-of-plant share rises from 60 % at 10 MW per year to 70 % at 1 GW per annum due to reduced stack production costs [212]. According to IRENA [16] and ETC [204], the electrolyzer stack accounts for 45 % and 55 % of AWE expenditures. The majority of this cost comes from producing the diaphragm/electrode containerize, and the BoP cost breakup is identical to that of PEMWE. In general, when capability or output levels improve, so does the stack's contribution. Fig. 7b shows cost breakdowns for various capacities, indicating cost reductions mostly connected with balance-of-plant. Assessing the financial implications of various stack fragments can be challenging due to the lack of an openly accessible inventory for electrolysis stacks, which will likely differ significantly between

manufacturers and production scales. According to NREL, AWE stacks cost USD 100 per kilowatt-hour with a 1 MW capacity, generating 10–20 units annually. Fig. 7c depicts differences in the literature concerning the cost contribution of several stack elements for PEMWE. The catalyst-coated membrane typically represents the highest cost (23–47 % of the total) because of the inclusion of iridium and platinum, whereas bipolar plates contribute a significant cost (9–51 %), based on the material used, with titanium plates costing more and gold-coated steel costing less [213].

8. Drivers of cost reduction

Lower prices will probably be due to a rapidly evolving and expanding market, specifically production scale-up, plant size growth, layout enhancements, and changing output to reduced worldwide locations.

8.1. Electrolyzer plant size

Although electrolyzers formerly worked at the kilowatt scale for several decades, the quantity of particular electrolyzer initiatives has grown substantially over the past ten years as production supply chains have matured. From 2010 to 2017, AWE systems expanded in size from 120 kW to an average of 2 MW, while PEMWE systems grew from 10 kW to 2.9 MW [203]. Projects are projected to grow by three orders of magnitude over the next decade, with a sharp rise in capacity from 1 to 5 MW in 2020 to 30–300 MW by 2025 (Fig. 8) [202,203]. The rise in plant size is projected to lower system costs via economies of scale.

8.2. Manufacturing scale-up

Expanding production processes allows for substantial savings in costs. Comparable to the benefits associated with expanding plant size, economies of scale during production may substantially decrease costs such as the usage of energy, needed materials, and labor, aided by rising automation and learning rates. In the case of electrolyzers, a transition from manual assembly and connection operations to large quantities production techniques like LASER cutting, plastic injection molding, and 3D printing offers the potential to reduce costs. Production knowledge is useful in reducing design risks and using finer margins, such as fewer needs for materials, to maximize costs, productivity, and lives.

8.3. Design improvements

Various technological innovations suited to each kind of electrolyzer have the potential to improve (η) or cut expenses associated with electrolyzer stacks. Refined mixed metal oxide catalysts enable AWEs to achieve current density increases ranging from 0.2 to 0.4 to 0.6 A cm^2 [33]. Operating at greater temperatures could boost performance by using stronger electrodes and electrolytes, but zero-gap designs remove the gap between electrodes and reduce electrolyte resistance and bubble formation. PEMWEs can achieve higher current densities (ranging from 0.6 to 2 to 43 A cm^2) by improving electrode design, catalytic coating, and using thin membranes. Thinner coatings might decrease dependence on iridium and platinum, potentially saving money.

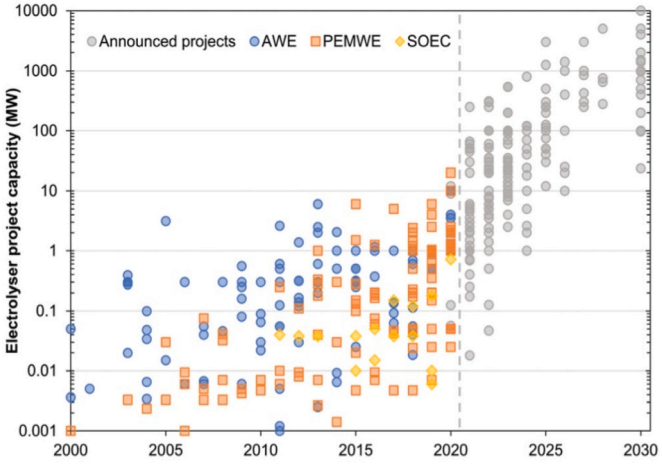


Fig: 8. Operational electrolysis plant size over the past 20 years and announced plans to develop during the next decade adapted from [26,202,203].

Furthermore, substituting titanium in bipolar plates and permeable transport layers that have elevated conductivity/stable coatings on inexpensive components such as steel could assist in lowering costs. The rectifier, which converts AC to DC, makes up a major percentage of capex and might be reduced by employing a DC source that just requires a DC/DC converter. Capex reductions in SOEC are possible by lowering the temperatures of operation to below 450 °C, thereby lowering electrode polarization resistance, eliminating the requirement for elevated temperatures exotic materials and allowing the use of lower-cost materials like stainless steel. While SOECs are still in the beginning stages of expansion, establishing lifetimes and improving cell and stack layouts is important. Fig. 9 depicts the possible combined cost savings related to design developments, higher plant scale, and production scale-up using a PEMWE system as a case study. The most substantial benefits come from manufacturing economies of scale, which increase output from 10 to 100 units per year, potentially lowering total expenses from billions to hundreds of dollars per kilowatt.

8.4. Shifting production centers

Another element contributing to projected cost savings is the shift of manufacturing from Western countries (mainly Europe and America) to China. This is consistent with the pattern seen in other environmentally friendly approaches, such as the quick drop in the price of solar PV panels after manufacturing shifted from Germany and the United States to China. The BNEF suggests three explanations for China's cheaper costs: lower raw material and labor costs, higher plant utilization rates, and lower R&D and marketing costs. Some argue that manufacturing quality, specifically low longevity and dependability, plays an impact. According to BNEF [211], AWEs produced in China went on sale for $200 per kilowatt in 2019, representing an 83 % decrease compared to Western-made installations at the time. Furthermore, the lessons learned from the COVID-19 epidemic may result in a rebirth of economic activities in Europe. While the IEA puts AWE costs at $500 per kilowatt, BNEF predicts that Western manufacturers' costs will exceed those of Chinese manufacturers in the next decade. Failing to improve cost competitiveness may result in Western manufacturers losing their share of the market and potentially facing economic difficulties. Agora argues that broad innovation assistance at the EU level is critical to the achievement of electrolysis production in Europe [219]. Table 5 summarizes the European Union, Australia, Japan, United States, and Canada strategies and policies toward green hydrogen production.

9. Future directions

There is an implicit belief in academia that a good catalyst is also a good electrode, so the main strategy for academia to reduce costs is to develop low-noble-metal catalysts or even noble metal-free catalysts. So far, noble metals account for only about 10 % of the overall cost of industrial electrolyzers [228]. Reducing noble metals is unquestionably an important task for the development of PEMWE, but it is difficult to accomplish through catalyst research alone. One important reason is the use of cutting-edge low-catalyst electrodes. For example, at a high current density of 4 A cm^{-2}, catalyst utilization in the ACL can be as low as 2.4 % for 3 M's nanostructured thin film (NSTF) electrode, while FuGe MEA is 30 % [229]. Low catalytic utilization is caused by a complex

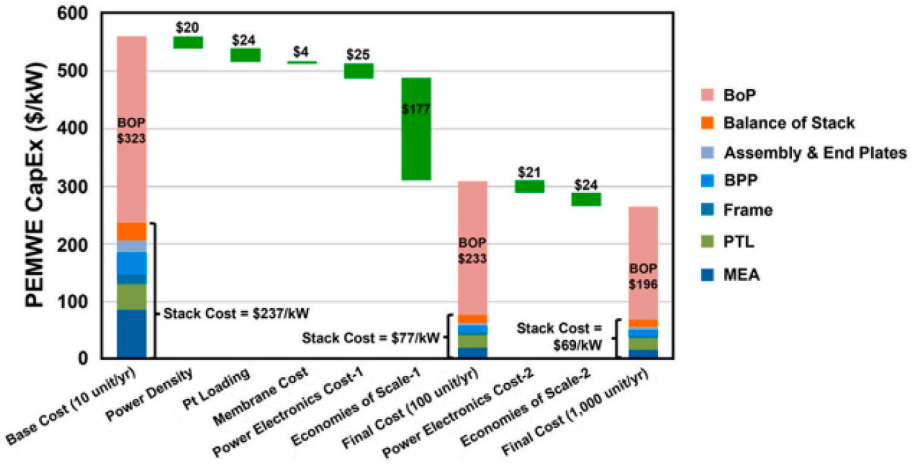


Fig: 9. Cost reductions for PEMWE systems in the future at various production levels adapted from [220].

Table 5
Compression of European union, Australia, Japan, United States and Canada strategies and policies toward green H_2 production [221–227].

Criteria	European union H_2 strategy	Green H_2 projects strategy in Australia	Japan H_2 society roadmap	Green H_2 in the US and Canada strategy
Targets	• 6 GW of electrolyzers by 2024, 40 GW by 2030 • 10 million metric tons of renewable H_2 by 2030	• Asian Renewable Energy Hub: 26 GW capacity • Tasmania: Global producer of green H_2 by 2030	• 200,000 FCEVs and 320 hydrogen refuelling stations by 2025 • Green H_2 cost reduction to ¥30 per Nm^3 by 2030	• U.S.: $1 per kg clean H_2 within a decade (H_2 energy Earths hot) • Canada: Top three global H_2 producer by 2050
R&D	• Clean H_2 partnership • Focus on cost reduction and (η) improvements	• Supported by Australian renewable energy agency (ARENA) • Woodside energy HyNet Project	• Collaboration between industry, research institutions, and government • R&D in electrolysis, H_2 storage, and transportation	• U.S. H_2 program-H_2@Scale initiative • Canada: Investment through various funding programs
Infrastructure Development	• European H_2 backbone • Retrofit existing natural gas pipelines and construct new pipelines and storage facilities	• AREH: Export to domestic and international markets • HyPSA: Distribution to local homes and businesses	• Construction of H_2 refuelling stations • Development of hydrogen pipelines and import facilities	• U.S.: H_2@Scale initiative • Canada: H_2 production facilities, distribution networks, and refuelling stations
Regulatory and Policy Frameworks	• Standards for renewable and low-carbon H_2 • Guarantees of origin-incorporating green H_2 into the EU ETS	• Tasmania renewable H_2 action plan • Renewable H_2 fund	• Subsidies for FCEVs and H_2 refuelling stations • Tax incentives for H_2 related investments • Inclusion of H_2 in Japan energy mix	• U.S.: DOE H_2 program and H_2 energy Earths hot • Canada: Strategic innovation fund, clean fuel fund, and low carbon economy fund
International Cooperation	• Partnerships with North Africa for green H_2 production • Development of global regulatory framework and standards	• Asian renewable energy hub: Export-oriented project	• H_2 energy ministerial meeting (HEM) • International partnership for H_2 and fuel cells in the economy (IPHE) • Bilateral partnerships with countries like Australia and Norway	• U.S.: International cooperation through DOE initiatives • Canada: Emphasis on international collaboration and export market development

balance between proton/electron and water/gas transport in ACL. As a result, it is necessary to increase research investment in ACL to increase catalyst utilization, which could reduce noble metals while also improving performance more efficiently. Anodes face significantly more technical challenges than cathodes, whose fabrication is well-established in the industry. Unfortunately, regular OER catalysts, primarily IrO_2-based nanoparticles, exhibit weak ionomer adsorption, resulting in low-viscosity catalyst inks that are difficult to deploy in CCM production lines [230]. As a result, instead of focusing solely on improving activity in academia, OER catalysts must be specifically optimized in industry for enhanced ionomer sorption. As a result, a catalyst must overcome a number of challenges to meet industrial requirements.

However, very few electrolyzers have operated for such a long time, and new technologies emerge frequently. The evaluation of real lifetime at such a timescale is nearly impossible for the academic community, which typically only supports stability testing for hundreds to thousands of hours [231]. Rather than stating stable performance by displaying an aging curve in most literature, the industry seeks to understand why and how materials degrade in electrolyzers. Generally, academic research to improve performance focuses on chemical methods, such as designing novels and more active catalysts. However, industry experience has shown that many performance and lifetime limitations stem from physical factors such as stress distribution on CCM. It was recently demonstrated that optimizing the interface with a stress distribution improved the performance and stability of electrolyzers [232]. Another example is the degradation of the anode catalyst layer. Heat and water continuously influence the microstructure of ACL due to the swelling, creeping, and migration properties of polymer ionomer binders [233]. Regardless of the fact that the majority of the literature discusses chemical degradation processes, industry has discovered that one major degradation mechanism is the collapse of the physical structure of the

anode catalyst layer. Unfortunately, physical processes have received far less attention from academia [234].

From the perspective of academia, the commercialization process entails almost a one-way scaling up of materials or technologies. However, one-way scaling may result in the loss of significant cost or performance advantages. For example, in larger electrolytes, the one-way scale-up strategy can frequently result in significant performance loss. This is since heat, mass, electron, and proton transfer all differ significantly in larger electrolyzers, whereas they have little influence in laboratory-scale cells [235]. Furthermore, to the academic community's drive for enhanced (η) through the development of novel materials or methods, it would be more practical to identify fundamental structure-function relationships to guide the development of commercial electrolyzers [26].

The perfect electrolyzer should operate efficiently at both partial load and overload, which is critical for PEMWE to take advantage of low-cost fluctuating electricity. Instead, an economically sound approach is to reduce electrolyzer capacity by increasing the upper limit of PEMWE overload (which is currently around 120 %). In addition to lowering material costs, increasing the current density is a cost-effective measure for lowering overall PEMWE expenses. Despite the global installation of hundreds of megawatts of PEMWE, there is still insufficient data on industrial PEMWE operating in realistic fluctuating power conditions. Therefore, academic research requires more detailed industry data feedback to effectively address real-world research problems. A further major problem is how fluctuating power input affects the lifespan of materials and entire electrolyzers [236]. Under dynamic operation, accelerated catalyst dissolution was noticed in liquid electrolyte tests [237], while PEMWE devices exhibit improved durability. As a result, it is difficult to reach a definitive conclusion about the effect of dynamic operation on PEMWE lifetime. The primary cause of the increased dissolution rate is catalyst surface reconstruction, which can be reduced by controlling the lowest possible potential during operation. Furthermore, it is important to note that directly extrapolating dissolution rates from liquid electrolyte tests to device conditions may result in an overestimation of catalyst degradation, as the acidic environment differs significantly between the two systems. There is an urgent need to develop a pragmatic testing methodology that aligns fundamental research with industry demands.

In conclusion, bridging the gap between academic research and industry is critical for advancing the promising PEMWE technology. Furthermore, the industry must provide comprehensive support, both in terms of technical expertise and industrial resources, for the development of new materials and technologies derived from academic research. Collaboration between academia and industry is critical for driving progress and ensuring the effective execution of innovative solutions in the field.

10. Conclusions

Hydrogen is poised to play a crucial role in the future's zero-emission energy system. Electrolysis stands out as a key technology for generating energy carriers, shaping the transition from fossil fuels to a zero-carbon energy system. Among the various electrolysis systems, PEMWE stands out for its reliability in producing hydrogen from renewable energy sources. Despite being in its early stages, significant enhancements are essential to augment durability, performance, and efficiency. This review offers a concise overview of major water electrolysis systems, exploring their techno-commercial prospects, market dynamics, and hydrogen production costs. The focus extends to understanding PEMWE degradation mechanisms, addressing challenges, and exploring potential strategies for improving cell components like membranes, gas diffusion layers, and bipolar plates in PEM water electrolysis. The aim is to present a comprehensive and realistic perspective on enhancing PEMWE (η) through diverse research platforms. The review also delves into notable achievements, identifies research gaps, and discusses avenues for reducing costs, scaling up, and commercializing PEM water electrolysis. Future studies should encompass the trajectory, efficiency, durability, and cost of PEMWE to contribute meaningfully to this field.

CRediT authorship contribution statement

Fazil Qureshi: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Mohammad Asif:** Writing – review & editing, Visualization, Formal analysis. **Abuzar Khan:** Writing – review & editing, Validation, Conceptualization. **Arafat Husain:** Writing – review & editing, Software, Conceptualization. **Hamad Aldawsari:** Writing – review & editing, Software. **Mohammad Yusuf:** Writing – review & editing, Software, Data curation. **Satish Kumar Verma:** Writing – review & editing, Methodology. **Mohd Yusuf Khan:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mohd Yusuf Khan reports was provided by King Fahd University of Petroleum & Minerals. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to acknowledge the support provided by the Hydrogen Future Consortium for project number "H2FC2302" under the Deanship of Research at King Fahd University of Petroleum and Minerals (KFUPM).

References

[1] Maheswari P, Haider MB, Yusuf M, Klemeš JJ, Bokhari A, Beg M, et al. A review on latest trends in cleaner biodiesel production: role of feedstock, production methods, and catalysts. J Clean Prod 2022;355:131588. https://doi.org/10.1016/j.jclepro.2022.131588.

[2] Ishaq H, Dincer I, Crawford C. A review on hydrogen production and utilization: challenges and opportunities. Int J Hydrogen Energy 2021. https://doi.org/10.1016/j.ijhydene.2021.11.149.

[3] Qureshi F, Yusuf M, Kamyab H, Zaidi S, Junaid Khalil M, Arham Khan M, et al. Current trends in hydrogen production, storage and applications in India: a review. Sustain Energy Technol Assessments 2022;53:102677. https://doi.org/10.1016/j.seta.2022.102677.

[4] Qureshi F, Yusuf M, Ahmed S, Haq M, Alraih AM, Hidouri T, et al. Advancements in sorption-based materials for hydrogen storage and utilization: a comprehensive review. Energy 2024;309:132855. https://doi.org/10.1016/j.energy.2024.132855.

[5] Afzal A, Soudagar MEM, Belhocine A, Kareemullah M, Hossain N, Alshahrani S, et al. Thermal performance of compression ignition engine using high content biodiesels: a comparative study with Diesel Fuel. Sustain Times 2021;13:7688. https://doi.org/10.3390/su13147688.

[6] Qureshi F, Yusuf M, Ibrahim H, Kamyab H, Chelliapan S, Pham CQ, et al. Contemporary avenues of the Hydrogen industry: opportunities and challenges in the eco-friendly approach. Environ Res 2023;229:115963. https://doi.org/10.1016/j.envres.2023.115963.

[7] Qureshi F, Yusuf M, Arham Khan M, Ibrahim H, Ekeoma BC, Kamyab H, et al. A State-of-The-Art Review on the Latest trends in Hydrogen production, storage, and transportation techniques. Fuel 2023;340:127574. https://doi.org/10.1016/j.fuel.2023.127574.

[8] Qureshi F, Yusuf M, Tahir M, Haq M, Mohamed MMI, Kamyab H, et al. Renewable hydrogen production via biological and thermochemical routes: nanomaterials, economic analysis and challenges. Process Saf Environ Prot 2023;179:68–88. https://doi.org/10.1016/j.psep.2023.07.075.

[9] Yusuf M, Rosha P, Qureshi F, Ali FM, Ibrahim H. Recent avenues in the photocatalytic splitting of water for eco-friendly hydrogen production. Sustain Mater Technol 2025:e01332. https://doi.org/10.1016/j.susmat.2025.e01332.

[10] Nnabuife SG, Ugbeh-Johnson J, Okeke NE, Ogbonnaya C. Present and projected developments in hydrogen production: a technological review. Carbon Capture Sci Technol 2022;3:100042. https://doi.org/10.1016/j.ccst.2022.100042.

[11] Mohammadi A, Mehrpooya M. A comprehensive review on coupling different types of electrolyzer to renewable energy sources. Energy 2018;158:632–55. https://doi.org/10.1016/j.energy.2018.06.073.

[12] Gopinath M, Marimuthu R. A review on solar energy-based indirect water-splitting methods for hydrogen generation. Int J Hydrogen Energy 2022;47:37742–59. https://doi.org/10.1016/j.ijhydene.2022.08.297.
[13] Lim D, Lee B, Lee H, Byun M, Lim H. Projected cost analysis of hybrid methanol production from tri-reforming of methane integrated with various water electrolysis systems: technical and economic assessment. Renew Sustain Energy Rev 2022;155:111876. https://doi.org/10.1016/j.rser.2021.111876.
[14] Lim D, Lee B, Lee H, Byun M, Cho H-S, Cho W, et al. Impact of voltage degradation in water electrolyzers on sustainability of synthetic natural gas production: energy, economic, and environmental analysis. Energy Convers Manag 2021;245:114516. https://doi.org/10.1016/j.enconman.2021.114516.
[15] Lee B, Lim D, Lee H, Lim H. Which water electrolysis technology is appropriate?: critical insights of potential water electrolysis for green ammonia production. Renew Sustain Energy Rev 2021;143:110963. https://doi.org/10.1016/j.rser.2021.110963.
[16] IRENA. Green hydrogen cost reduction: scaling up electrolysers to meet the 1.5^{0}C climate goal. International Renewable Energy Agency 2020:105. Irena 2020.
[17] Shiva Kumar S, Lim H. An overview of water electrolysis technologies for green hydrogen production. Energy Rep 2022;8:13793–813. https://doi.org/10.1016/j.egyr.2022.10.127.
[18] Qureshi F, Ahmad F, Idrees M, Khan AA, Zaidi S. Simulation of methanol steam reforming process for the production of hydrogen. Indian Chem Eng 2021;63:99–116. https://doi.org/10.1080/00194506.2019.1689186.
[19] Bernt M, Hartig-Weiß A, Tovini MF, El-Sayed HA, Schramm C, Schröter J, et al. Current challenges in catalyst development for PEM water electrolyzers. Chemie Ing Tech 2020;92:31–9. https://doi.org/10.1002/cite.201900101.
[20] Ouimet RJ, Glenn JR, De Porcellinis D, Motz AR, Carmo M, Ayers KE. The role of electrocatalysts in the development of gigawatt-scale PEM electrolyzers. ACS Catal 2022;12:6159–71. https://doi.org/10.1021/acscatal.2c00570.
[21] Dang J, Zhang J, Deng X, Yang S, Liu B, Zhu X, et al. Hydrogen crossover measurement and durability assessment of high-pressure proton exchange membrane electrolyzer. J Power Sources 2023;563:232776. https://doi.org/10.1016/j.jpowsour.2023.232776.
[22] Li Y, Li H, Liu W, Zhu Q. Optimization of membrane thickness for proton exchange membrane electrolyzer considering hydrogen production efficiency and hydrogen permeation phenomenon. Appl Energy 2024;355:122233. https://doi.org/10.1016/j.apenergy.2023.122233.
[23] Kumar SS, Lim H. Recent advances in hydrogen production through proton exchange membrane water electrolysis – a review. Sustain Energy Fuels 2023;7:3560–83. https://doi.org/10.1039/D3SE00336A.
[24] Park JE, Kim J, Han J, Kim K, Park S, Kim S, et al. High-performance proton-exchange membrane water electrolysis using a sulfonated poly(arylene ether sulfone) membrane and ionomer. J Memb Sci 2021;620:118871. https://doi.org/10.1016/j.memsci.2020.118871.
[25] Hong SJ, Jung HY, Yoon SJ, Oh K-H, Oh S-G, Hong YT, et al. Constrained hydrocarbon-based ionomers in porous Poly(tetrafluoroethylene) supports for enhanced durability of polymer electrolyte membrane fuel cells and water electrolyzers. J Power Sources 2022;551:232221. https://doi.org/10.1016/j.jpowsour.2022.232221.
[26] Chatenet M, Pollet BG, Dekel DR, Dionigi F, Deseure J, Millet P, et al. Water electrolysis: from textbook knowledge to the latest scientific strategies and industrial developments. Chem Soc Rev 2022;51:4583–762. https://doi.org/10.1039/D0CS01079K.
[27] EIA. Use of hydrogen. U.S. Energy Information Administration (EIA); 2023. https://wwwEiaGov/Energyexplained/Hydrogen/Use-of-HydrogenPhp. [Accessed 17 February 2024]. https://www.eia.gov/energyexplained/hydrogen/use-of-hydrogen.php.
[28] Qureshi F, Yusuf M, Kamyab H, Vo DVN, Chelliapan S, Joo SW, et al. Latest eco-friendly avenues on hydrogen production towards a circular bioeconomy: currents challenges, innovative insights, and future perspectives. Renew Sustain Energy Rev 2022;168:112916. https://doi.org/10.1016/j.rser.2022.112916.
[29] Newborough M, Cooley G. Developments in the global hydrogen market: the spectrum of hydrogen colours. Fuel Cells Bull 2020;2020:16–22. https://doi.org/10.1016/S1464-2859(20)30546-0.
[30] Qureshi F, Yusuf M, Abdullah B. A brief review on hydrogen production to utilization techniques. Third Int Sustain Resil Conf Clim Chang 2021:66–71. https://doi.org/10.1109/IEEECONF53624.2021.9668127.
[31] Qureshi F, Yusuf M, Pasha AA, Khan HW, Imteyaz B, Irshad K. Sustainable and energy efficient hydrogen production via glycerol reforming techniques: a review. Int J Hydrogen Energy 2022;47:41397–420. https://doi.org/10.1016/j.ijhydene.2022.04.010.
[32] Hermesmann M, Grübel K, Scherotzki L, Müller TE. Promising pathways: the geographic and energetic potential of power-to-x technologies based on regeneratively obtained hydrogen. Renew Sustain Energy Rev 2021;138:110644. https://doi.org/10.1016/j.rser.2020.110644.
[33] Grigoriev SA, Fateev VN, Bessarabov DG, Millet P. Current status, research trends, and challenges in water electrolysis science and technology. Int J Hydrogen Energy 2020;45:26036–58. https://doi.org/10.1016/j.ijhydene.2020.03.109.
[34] Carmo M, Keeley GP, Holtz D, Grube T, Robinius M, Müller M, et al. PEM water electrolysis: innovative approaches towards catalyst separation, recovery and recycling. Int J Hydrogen Energy 2019;44:3450–5. https://doi.org/10.1016/j.ijhydene.2018.12.030.
[35] Zhang K, Liang X, Wang L, Sun K, Wang Y, Xie Z, et al. Status and perspectives of key materials for PEM electrolyzer. Nano Res Energy 2022;1:e9120032. https://doi.org/10.26599/NRE.2022.9120032.
[36] Bareiß K, de la Rua C, Möckl M, Hamacher T. Life cycle assessment of hydrogen from proton exchange membrane water electrolysis in future energy systems. Appl Energy 2019;237:862–72. https://doi.org/10.1016/j.apenergy.2019.01.001.
[37] Air Liquide starts up Cummins 20 MW PEM electrolyser in Canada. Fuel Cells Bull 2021;2021:12–3. https://doi.org/10.1016/S1464-2859(21)00099-7.
[38] Snieckus D. Hydrogen electrolyser market booms with "1,000-fold" growth in frame by 2040: Aurora. 2021.
[39] Woodward E, Oh CF. Enabling the European hydrogen economy. Aurora Energy Res 2021:1–29.
[40] Qureshi F, Tahir M. Photoelectrochemical water splitting with engineering aspects for hydrogen production: recent advances, strategies and challenges. Int J Hydrogen Energy 2024;69:760–76. https://doi.org/10.1016/j.ijhydene.2024.05.039.
[41] Zerga AY, Qureshi F, Tahir M. Recent developments of scalable photocatalytic reactors for green hydrogen production. ACS Symp Ser 2024:211–42. https://doi.org/10.1021/bk-2024-1467.ch010.
[42] Minke C, Suermann M, Bensmann B, Hanke-Rauschenbach R. Is iridium demand a potential bottleneck in the realization of large-scale PEM water electrolysis? Int J Hydrogen Energy 2021;46:23581–90. https://doi.org/10.1016/j.ijhydene.2021.04.174.
[43] Idriss H. Toward large-scale hydrogen production from water: what have we learned and what are the main research hurdles to cross for commercialization? Energy Technol 2021;9. https://doi.org/10.1002/ente.202000843.
[44] Grubert E. Water consumption from electrolytic hydrogen in a carbon-neutral US energy system. Clean Prod Lett 2023;4:100037. https://doi.org/10.1016/j.clpl.2023.100037.
[45] Winter LR, Cooper NJ, Lee B, Patel SK, Wang L, Elimelech M. Mining nontraditional water sources for a distributed hydrogen economy. Environ Sci Technol 2022;56:10577–85. https://doi.org/10.1021/acs.est.2c02439.
[46] Dresp S, Dionigi F, Klingenhof M, Strasser P. Direct electrolytic splitting of seawater: opportunities and challenges. ACS Energy Lett 2019;4:933–42. https://doi.org/10.1021/acsenergylett.9b00220.
[47] Dresp S, Dionigi F, Loos S, Ferreira de Araujo J, Spöri C, Gliech M, et al. Direct electrolytic splitting of seawater: activity, selectivity, degradation, and recovery studied from the molecular catalyst structure to the electrolyzer cell level. Adv Energy Mater 2018;8. https://doi.org/10.1002/aenm.201800338.
[48] Cohen Y. Advances in water desalination technologies, vol. 17. World Scientific; 2021. https://doi.org/10.1142/12009.
[49] Mustafa J, Al-Marzouqi AH, Ghasem N, El-Naas MH, Van der Bruggen B. Electrodialysis process for carbon dioxide capture coupled with salinity reduction: a statistical and quantitative investigation. Desalination 2023;548:116263. https://doi.org/10.1016/j.desal.2022.116263.
[50] Panagopoulos A, Haralambous K-J. Environmental impacts of desalination and brine treatment - challenges and mitigation measures. Mar Pollut Bull 2020;161:111773. https://doi.org/10.1016/j.marpolbul.2020.111773.
[51] Sharkh BA, Al-Amoudi AA, Farooque M, Fellows CM, Ihm S, Lee S, et al. Seawater desalination concentrate—a new frontier for sustainable mining of valuable minerals. Npj Clean Water 2022;5:9. https://doi.org/10.1038/s41545-022-00153-6.
[52] Tong W, Forster M, Dionigi F, Dresp S, Sadeghi Erami R, Strasser P, et al. Electrolysis of low-grade and saline surface water. Nat Energy 2020;5:367–77. https://doi.org/10.1038/s41560-020-0550-8.
[53] Khan MA, Al-Attas T, Roy S, Rahman MM, Ghaffour N, Thangadurai V, et al. Seawater electrolysis for hydrogen production: a solution looking for a problem? Energy Environ Sci 2021;14:4831–9. https://doi.org/10.1039/d1ee00870f.
[54] Guo J, Zheng Y, Hu Z, Zheng C, Mao J, Du K, et al. Direct seawater electrolysis by adjusting the local reaction environment of a catalyst. Nat Energy 2023. https://doi.org/10.1038/s41560-023-01195-x.
[55] Hausmann JN, Schlögl R, Menezes PW, Driess M. Is direct seawater splitting economically meaningful? Energy Environ Sci 2021;14:3679–85. https://doi.org/10.1039/D0EE03659E.
[56] Juárez-Casildo V, Cervantes I, Cervantes-Ortiz CA, González-Huerta R de G. Key aspects in quantifying massive solar hydrogen production: energy intermittence, water availability and electrolyzer technology. J Clean Prod 2022;371:133550. https://doi.org/10.1016/j.jclepro.2022.133550.
[57] Lebrouhi BE, Djoupo JJ, Lamrani B, Benabdelaziz K, Kousksou T. Global hydrogen development - a technological and geopolitical overview. Int J Hydrogen Energy 2022;47:7016–48. https://doi.org/10.1016/j.ijhydene.2021.12.076.
[58] Lee JM, Lee S, Kye DH, Park HJ, Park W, Shin J, et al. Environ-economic analysis of high-temperature steam electrolysis for decentralized hydrogen production. Energy Convers Manag 2022;266:115856. https://doi.org/10.1016/j.enconman.2022.115856.
[59] Arsad AZ, Hannan MA, Al-Shetwi AQ, Hossain MJ, Begum RA, Ker PJ, et al. Hydrogen electrolyser for sustainable energy production: a bibliometric analysis and future directions. Int J Hydrogen Energy 2023;48:4960–83. https://doi.org/10.1016/j.ijhydene.2022.11.023.
[60] Lim T, Kim J, Joo SH. Electrocatalysis of selective chlorine evolution reaction: fundamental understanding and catalyst design. J Electrochem Sci Technol 2023;14:105–19. https://doi.org/10.33961/jecst.2022.01032.
[61] Loomba S, Khan MW, Mahmood N. Seawater to green hydrogen: future of green energy. Chemelectrochem 2023;10. https://doi.org/10.1002/celc.202300471.
[62] Yu Z, Li Y, Martin-Diaconescu V, Simonelli L, Ruiz Esquius J, Amorim I, et al. Highly efficient and stable saline water electrolysis enabled by self-supported nickel-iron phosphosulfide nanotubes with heterointerfaces and under-

coordinated metal active sites. Adv Funct Mater 2022;32. https://doi.org/10.1002/adfm.202206138.
[63] Dresp S, Ngo Thanh T, Klingenhof M, Brückner S, Hauke P, Strasser P. Efficient direct seawater electrolysers using selective alkaline NiFe-LDH as OER catalyst in asymmetric electrolyte feeds. Energy Environ Sci 2020;13:1725–9. https://doi.org/10.1039/D0EE01125H.
[64] Tong W, Forster M, Dionigi F, Dresp S, Erami RS, Strasser P, et al. Publisher Correction: electrolysis of low-grade and saline surface water. Nat Energy 2021;6:935–6. https://doi.org/10.1038/s41560-021-00851-4.
[65] Lindquist GA, Xu Q, Oener SZ, Boettcher SW. Membrane electrolyzers for impure-water splitting. Joule 2020;4:2549–61. https://doi.org/10.1016/j.joule.2020.09.020.
[66] Sun F, Qin J, Wang Z, Yu M, Wu X, Sun X, et al. Energy-saving hydrogen production by chlorine-free hybrid seawater splitting coupling hydrazine degradation. Nat Commun 2021;12:4182. https://doi.org/10.1038/s41467-021-24529-3.
[67] Xie H, Zhao Z, Liu T, Wu Y, Lan C, Jiang W, et al. A membrane-based seawater electrolyser for hydrogen generation. Nature 2022;612:673–8. https://doi.org/10.1038/s41586-022-05379-5.
[68] Loomba S, Khan MW, Haris M, Mousavi SM, Zavabeti A, Xu K, et al. Nitrogen-doped porous nickel molybdenum phosphide sheets for efficient seawater splitting. Small 2023;19. https://doi.org/10.1002/smll.202207310.
[69] Shehata N, Egirani D, Olabi AG, Inayat A, Abdelkareem MA, Chae K-J, et al. Membrane-based water and wastewater treatment technologies: issues, current trends, challenges, and role in achieving sustainable development goals, and circular economy. Chemosphere 2023;320:137993. https://doi.org/10.1016/j.chemosphere.2023.137993.
[70] van Vliet MTH, Jones ER, Flörke M, Franssen WHP, Hanasaki N, Wada Y, et al. Global water scarcity including surface water quality and expansions of clean water technologies. Environ Res Lett 2021;16:024020. https://doi.org/10.1088/1748-9326/abbfc3.
[71] Elsaid K, Taha Sayed E, Yousef BAA, Kamal Hussien Rabaia M, Ali Abdelkareem M, Olabi AG. Recent progress on the utilization of waste heat for desalination: a review. Energy Convers Manag 2020;221:113105. https://doi.org/10.1016/j.enconman.2020.113105.
[72] Hesari F, Salimnezhad F, Khoshgoftar Manesh MH, Morad MR. A novel configuration for low-grade heat-driven desalination based on cascade MED. Energy 2021;229:120657. https://doi.org/10.1016/j.energy.2021.120657.
[73] Temiz M, Dincer I. A unique ocean and solar based multigenerational system with hydrogen production and thermal energy storage for Arctic communities. Energy 2022;239:122126. https://doi.org/10.1016/j.energy.2021.122126.
[74] Xu Y-P, Lin Z-H, Ma T-X, She C, Xing S-M, Qi L-Y, et al. Optimization of a biomass-driven Rankine cycle integrated with multi-effect desalination, and solid oxide electrolyzer for power, hydrogen, and freshwater production. Desalination 2022;525:115486. https://doi.org/10.1016/j.desal.2021.115486.
[75] Khan MA, Al-Attas T, Roy S, Rahman MM, Ghaffour N, Thangadurai V, et al. Seawater electrolysis for hydrogen production: a solution looking for a problem? Energy Environ Sci 2021;14:4831–9. https://doi.org/10.1039/D1EE00870F.
[76] Abusultan AAM, Wood JA, Sainio T, Kemperman AJB, van der Meer WGJ. A hybrid process combining ion exchange resin and bipolar membrane electrodialysis for reverse osmosis remineralization. Desalination 2024;573:117209. https://doi.org/10.1016/j.desal.2023.117209.
[77] Eke J, Yusuf A, Giwa A, Sodiq A. The global status of desalination: an assessment of current desalination technologies, plants and capacity. Desalination 2020;495:114633. https://doi.org/10.1016/j.desal.2020.114633.
[78] Alsarayreh AA, Al-Obaidi MA, Farag SK, Patel R, Mujtaba IM. Performance evaluation of a medium-scale industrial reverse osmosis brackish water desalination plant with different brands of membranes. A simulation study. Desalination 2021;503:114927. https://doi.org/10.1016/j.desal.2020.114927.
[79] Nthunya LN, Bopape MF, Mahlangu OT, Mamba BB, Van der Bruggen B, Quist-Jensen CA, et al. Fouling, performance and cost analysis of membrane-based water desalination technologies: a critical review. J Environ Manage 2022;301:113922. https://doi.org/10.1016/j.jenvman.2021.113922.
[80] Saleem H, Zaidi SJ, Ismail AF, Goh PS, Vinu A. Recent advances in the application of carbon nitrides for advanced water treatment and desalination technology. Desalination 2022;542:116061. https://doi.org/10.1016/j.desal.2022.116061.
[81] Pratap B, Kumar S, Nand S, Azad I, Bharagava RN, Romanholo Ferreira LF, et al. Wastewater generation and treatment by various eco-friendly technologies: possible health hazards and further reuse for environmental safety. Chemosphere 2023;313:137547. https://doi.org/10.1016/j.chemosphere.2022.137547.
[82] Kumar P, Date A, Mahmood N, Kumar Das R, Shabani B. Freshwater supply for hydrogen production: an underestimated challenge. Int J Hydrogen Energy 2024;78:202–17. https://doi.org/10.1016/j.ijhydene.2024.06.257.
[83] Gholamian E, Mehr AS, Yari M, Carton JG. Dynamic simulation and techno-economic assessment of hydrogen utilization in dual fuel (Hydrogen/biogas) micro gas turbine systems for a wastewater treatment plant. Process Saf Environ Prot 2023;169:220–37. https://doi.org/10.1016/j.psep.2022.10.045.
[84] Musa Ardo F, Wei Lim J, Ramli A, Kee Lam M, Kiatkittipong W, Alaaeldin Abdelfattah E, et al. A review in redressing challenges to produce sustainable hydrogen from microalgae for aviation industry. Fuel 2022;330:125646. https://doi.org/10.1016/j.fuel.2022.125646.
[85] Wang S, Yang H, Shi Z, Zaini IN, Wen Y, Jiang J, et al. Renewable hydrogen production from the organic fraction of municipal solid waste through a novel carbon-negative process concept. Energy 2022;252:124056. https://doi.org/10.1016/j.energy.2022.124056.
[86] Cvetković SM, Radoičić TK, Novaković JG, Kovačević V, Lopičić ZR, Adamović V, et al. Renewable hydrogen production perspective in Serbia via biogas generated from food processing wastewaters. J Clean Prod 2022;363:132142. https://doi.org/10.1016/j.jclepro.2022.132142.
[87] Qyyum MA, Ismail S, Ni SQ, Ihsanullah I, Ahmad R, Khan A, et al. Harvesting biohydrogen from industrial wastewater: production potential, pilot-scale bioreactors, commercialization status, techno-economics, and policy analysis. J Clean Prod 2022;340:130809. https://doi.org/10.1016/j.jclepro.2022.130809.
[88] Chaurasia AK, Mondal P. Enhancing biohydrogen production from sugar industry wastewater using Ni, Ni–Co and Ni–Co–P electrodeposits as cathodes in microbial electrolysis cells. Chemosphere 2022;286:131728. https://doi.org/10.1016/j.chemosphere.2021.131728.
[89] Becker H, Murawski J, Shinde DV, Stephens IEL, Hinds G, Smith G. Impact of impurities on water electrolysis: a review. Sustain Energy Fuels 2023;7:1565–603. https://doi.org/10.1039/D2SE01517J.
[90] Wallnöfer-Ogris E, Grimmer I, Ranz M, Höglinger M, Kartusch S, Rauh J, et al. A review on understanding and identifying degradation mechanisms in PEM water electrolysis cells: insights for stack application, development, and research. Int J Hydrogen Energy 2024;65:381–97. https://doi.org/10.1016/j.ijhydene.2024.04.017.
[91] Wallnöfer-Ogris E, Poimer F, Köll R, Macherhammer M-G, Trattner A. Main degradation mechanisms of polymer electrolyte membrane fuel cell stacks – mechanisms, influencing factors, consequences, and mitigation strategies. Int J Hydrogen Energy 2024;50:1159–82. https://doi.org/10.1016/j.ijhydene.2023.06.215.
[92] Liang J, Wang B, Yin Y, Jiao K. Experimental investigation of operating characteristics of proton exchange membrane fuel cell with different anode strategies based on the segmented cell. Int J Green Energy 2024;21:798–815. https://doi.org/10.1080/15435075.2023.2219730.
[93] Lin Z, Wang T, Li Q. Designing active and stable Ir-based catalysts for the acidic oxygen evolution reaction. Ind Chem Mater 2023;1:299–311. https://doi.org/10.1039/D3IM00070B.
[94] Kasian O, Grote J, Geiger S, Cherevko S, Mayrhofer KJJ. The common intermediates of oxygen evolution and dissolution reactions during water electrolysis on iridium. Angew Chemie Int Ed 2018;57:2488–91. https://doi.org/10.1002/anie.201709652.
[95] Yu H, Bonville L, Jankovic J, Maric R. Microscopic insights on the degradation of a PEM water electrolyzer with ultra-low catalyst loading. Appl Catal B Environ 2020;260:118194. https://doi.org/10.1016/j.apcatb.2019.118194.
[96] Milosevic M, Böhm T, Körner A, Bierling M, Winkelmann L, Ehelebe K, et al. In search of lost iridium: quantification of anode catalyst layer dissolution in proton exchange membrane water electrolyzers. ACS Energy Lett 2023;8:2682–8. https://doi.org/10.1021/acsenergylett.3c00193.
[97] Zeng Z, Ouimet R, Bonville L, Niedzwiecki A, Capuano C, Ayers K, et al. Degradation mechanisms in advanced MEAs for PEM water electrolyzers fabricated by reactive spray deposition technology. J Electrochem Soc 2022;169:054536. https://doi.org/10.1149/1945-7111/ac7170.
[98] Li N, Araya SS, Cui X, Kær SK. The effects of cationic impurities on the performance of proton exchange membrane water electrolyzer. J Power Sources 2020;473:228617. https://doi.org/10.1016/j.jpowsour.2020.228617.
[99] Frensch SH, Fouda-Onana F, Serre G, Thoby D, Araya SS, Kær SK. Influence of the operation mode on PEM water electrolysis degradation. Int J Hydrogen Energy 2019;44:29889–98. https://doi.org/10.1016/j.ijhydene.2019.09.169.
[100] Moriau L, Smiljanić M, Lončar A, Hodnik N. Supported iridium-based oxygen evolution reaction electrocatalysts - recent developments. ChemCatChem 2022;14. https://doi.org/10.1002/cctc.202200586.
[101] Liu R-T, Xu Z-L, Li F-M, Chen F-Y, Yu J-Y, Yan Y, et al. Recent advances in proton exchange membrane water electrolysis. Chem Soc Rev 2023;52:5652–83. https://doi.org/10.1039/D2CS00681B.
[102] Tomić AZ, Pivac I, Barbir F. A review of testing procedures for proton exchange membrane electrolyzer degradation. J Power Sources 2023;557:232569. https://doi.org/10.1016/j.jpowsour.2022.232569.
[103] Frensch SH, Serre G, Fouda-Onana F, Jensen HC, Christensen ML, Araya SS, et al. Impact of iron and hydrogen peroxide on membrane degradation for polymer electrolyte membrane water electrolysis: computational and experimental investigation on fluoride emission. J Power Sources 2019;420:54–62. https://doi.org/10.1016/j.jpowsour.2019.02.076.
[104] Qureshi F, Asif M, Khan A, Aldawsari H, Yusuf M, Khan MY. Green hydrogen production from non-traditional water sources: a sustainable energy solution with hydrogen storage and distribution. Chem Rec 2024:1–22. https://doi.org/10.1002/tcr.202400080.
[105] Song Y, Chen H, Wang X, Weng C, Zou K, Wang C, et al. Engineering Ir-based catalysts for high current density applications in proton exchange membrane water electrolyzers. Energy Environ Sci 2024. https://doi.org/10.1039/D4EE03541K.
[106] Siracusano S, Trocino S, Briguglio N, Pantò F, Aricò AS. Analysis of performance degradation during steady-state and load-thermal cycles of proton exchange membrane water electrolysis cells. J Power Sources 2020;468:228390. https://doi.org/10.1016/j.jpowsour.2020.228390.
[107] Krenz T, Weiland O, Trinke P, Helmers L, Eckert C, Bensmann B, et al. Temperature and performance inhomogeneities in PEM electrolysis stacks with industrial scale cells. J Electrochem Soc 2023;170:044508. https://doi.org/10.1149/1945-7111/accb68.
[108] Chandesris M, Médeau V, Guillet N, Chelghoum S, Thoby D, Fouda-Onana F. Membrane degradation in PEM water electrolyzer: numerical modeling and experimental evidence of the influence of temperature and current density. Int J

Hydrogen Energy 2015;40:1353–66. https://doi.org/10.1016/j.ijhydene.2014.11.111.
[109] Wang K, Feng Y, Xiao F, Zhang T, Wang Z, Ye F, et al. Operando analysis of through-plane interlayer temperatures in the PEM electrolyzer cell under various operating conditions. Appl Energy 2023;348:121588. https://doi.org/10.1016/j.apenergy.2023.121588.
[110] Liu C, Shviro M, Gago AS, Zaccarine SF, Bender G, Gazdzicki P, et al. Exploring the interface of skin-layered titanium fibers for electrochemical water splitting. Adv Energy Mater 2021;11. https://doi.org/10.1002/aenm.202002926.
[111] Prestat M. Corrosion of structural components of proton exchange membrane water electrolyzer anodes: a review. J Power Sources 2023;556:232469. https://doi.org/10.1016/j.jpowsour.2022.232469.
[112] Ren P, Pei P, Chen D, Zhang L, Li Y, Song X, et al. Corrosion of metallic bipolar plates accelerated by operating conditions in a simulated PEM fuel cell cathode environment. Renew Energy 2022;194:1277–87. https://doi.org/10.1016/j.renene.2022.05.153.
[113] Tawancy HM. On the hydrogen embrittlement of commercially pure alpha titanium: an example from the petrochemical industry. J Mater Eng Perform 2017;26:504–13. https://doi.org/10.1007/s11665-016-2464-y.
[114] Liu Y, Huang S, Wang D, Zhang H, Shan D, Peng S, et al. Modifying Ti-based gas diffusion layer passivation for polymer electrolyte membrane water electrolysis via electrochemical nitridation. ACS Appl Mater Interfaces 2022;14:15728–35. https://doi.org/10.1021/acsami.1c22639.
[115] Tijani AS, Ghani MFA, Rahim AHA, Muritala IK, Binti Mazlan FA. Electrochemical characteristics of (PEM) electrolyzer under influence of charge transfer coefficient. Int J Hydrogen Energy 2019;44:27177–89. https://doi.org/10.1016/j.ijhydene.2019.08.188.
[116] Deloitte. Investing in hydrogen: ready, set, net-zero. 2020.
[117] Wakayama H, Yamazaki K. Low-cost bipolar plates of Ti 4 O 7 -coated Ti for water electrolysis with polymer electrolyte membranes. ACS Omega 2021;6:4161–6. https://doi.org/10.1021/acsomega.0c04786.
[118] Becker H, Dickinson EJF, Lu X, Bexell U, Proch S, Moffatt C, et al. Assessing potential profiles in water electrolysers to minimise titanium use. Energy Environ Sci 2022;15:2508–18. https://doi.org/10.1039/D2EE00876A.
[119] Shin H-S, Oh BS. Water transport according to temperature and current in PEM water electrolyzer. Int J Hydrogen Energy 2020;45:56–63. https://doi.org/10.1016/j.ijhydene.2019.10.209.
[120] Panchenko U, Arlt T, Manke I, Müller M, Stolten D, Lehnert W. Synchrotron radiography for a proton exchange membrane (PEM) electrolyzer. Fuel Cells 2020;20:300–6. https://doi.org/10.1002/fuce.201900055.
[121] Toghyani S, Afshari E, Baniasadi E, Atyabi SA, Naterer GF. Thermal and electrochemical performance assessment of a high temperature PEM electrolyzer. Energy 2018;152:237–46. https://doi.org/10.1016/j.energy.2018.03.140.
[122] Choi Y, Lee W, Na Y. Effect of gravity and various operating conditions on proton exchange membrane water electrolysis cell performance. Membranes 2021;11:822. https://doi.org/10.3390/membranes11110822.
[123] Moreno Soriano R, Rojas N, Nieto E, de Guadalupe González-Huerta R, Sandoval-Pineda JM. Influence of the gasket materials on the clamping pressure distribution in a PEM water electrolyzer: bolt torques and operation mode in pre-conditioning. Int J Hydrogen Energy 2021;46:25944–53. https://doi.org/10.1016/j.ijhydene.2021.03.076.
[124] Ogumerem GS, Pistikopoulos EN. Parametric optimization and control for a smart proton exchange membrane water electrolysis (PEMWE) system. J Process Control 2020;91:37–49. https://doi.org/10.1016/j.jprocont.2020.05.002.
[125] Tjarks G, Gibelhaus A, Lanzerath F, Müller M, Bardow A, Stolten D. Energetically-optimal PEM electrolyzer pressure in power-to-gas plants. Appl Energy 2018;218:192–8. https://doi.org/10.1016/j.apenergy.2018.02.155.
[126] Scheepers F, Stähler M, Stähler A, Rauls E, Müller M, Carmo M, et al. Improving the efficiency of PEM electrolyzers through membrane-specific pressure optimization. Energies 2020;13:612. https://doi.org/10.3390/en13030612.
[127] Afshari E, Khodabakhsh S, Jahantigh N, Toghyani S. Performance assessment of gas crossover phenomenon and water transport mechanism in high pressure PEM electrolyzer. Int J Hydrogen Energy 2021;46:11029–40. https://doi.org/10.1016/j.ijhydene.2020.10.180.
[128] Li X, Yao Y, Tian Y, Jia J, Ma W, Yan X, et al. Recent advances in key components of proton exchange membrane water electrolysers. Mater Chem Front 2024;8:2493–510. https://doi.org/10.1039/D4QM00086B.
[129] Ji W, Wang S, Sun Y, Lv H, Shen X, Zhang C. Research on the influence of collector microstructure on the performance of PEM electrolyzer. World Electr Veh J 2021;12:165. https://doi.org/10.3390/wevj12040165.
[130] Zhang J, Chen S, Bai H, Lu S, Xiang Y, Jiang SP. Effects of phosphotungstic acid on performance of phosphoric acid doped polyethersulfone-polyvinylpyrrolidone membranes for high temperature fuel cells. Int J Hydrogen Energy 2021;46:11104–14. https://doi.org/10.1016/j.ijhydene.2020.07.082.
[131] Barati S, Mehdipourghazi M, Abdollahi M, Hooshyari K, Khoshandam B. Preparation, characterization and proton transport of new porous nanocomposite membranes based on polybenzimidazole, lignin and <scp> TiO $_2$ </scp> nanoparticles for high temperature PEM fuel cells. Int J Energy Res 2021;45:20057–72. https://doi.org/10.1002/er.7083.
[132] Siracusano S, Pantò F, Tonella S, Oldani C, Aricò AS. Reinforced short-side-chain Aquivion® membrane for proton exchange membrane water electrolysis. Int J Hydrogen Energy 2022;47:15557–70. https://doi.org/10.1016/j.ijhydene.2022.03.061.
[133] Liu F, Wang S, Li J, Wang X, Yong Z, Cui Y, et al. Novel double cross-linked membrane based on poly (ionic liquid) and polybenzimidazole for high-temperature proton exchange membrane fuel cells. J Power Sources 2021;515:230637. https://doi.org/10.1016/j.jpowsour.2021.230637.
[134] Huang Z, Ying Z, Li R, Sun W, Zhang H, Wang Z, et al. Sub-two-micron ultrathin proton exchange membrane with reinforced mechanical strength. Polymer (Guildf) 2023;272:125829. https://doi.org/10.1016/j.polymer.2023.125829.
[135] Yuan Z, Tang J, Chen D, Li Y, Hong Z, He X. Membranes for hydrogen rainbow toward industrial decarbonization: status, challenges and perspectives from materials to processes. Chem Eng J 2023;470:144328. https://doi.org/10.1016/j.cej.2023.144328.
[136] Altaf CT, Colak TO, Karagoz E, Kurt M, Sankir ND, Sankir M. A review of the recent advances in composite membranes for hydrogen generation technologies. ACS Omega 2024;9:23138–54. https://doi.org/10.1021/acsomega.4c00152.
[137] Makhsoos A, Kandidayeni M, Pollet BG, Boulon L. A perspective on increasing the efficiency of proton exchange membrane water electrolyzers– a review. Int J Hydrogen Energy 2023;48:15341–70. https://doi.org/10.1016/j.ijhydene.2023.01.048.
[138] Lv H, Wang S, Hao C, Zhou W, Li J, Xue M, et al. Oxygen-deficient Ti 0.9 Nb 0.1 O 2-x as an efficient anodic catalyst support for PEM water electrolyzer. ChemCatChem 2019;11:2511–9. https://doi.org/10.1002/cctc.201900090.
[139] Zhang B, Fan L, Ambre RB, Liu T, Meng Q, Timmer BJJ, et al. Advancing proton exchange membrane electrolyzers with molecular catalysts. Joule 2020;4:1408–44. https://doi.org/10.1016/j.joule.2020.06.001.
[140] King LA, Hubert MA, Capuano C, Manco J, Danilovic N, Valle E, et al. A non-precious metal hydrogen catalyst in a commercial polymer electrolyte membrane electrolyser. Nat Nanotechnol 2019;14:1071–4. https://doi.org/10.1038/s41565-019-0550-7.
[141] Holzapfel PKR, Bühler M, Escalera-López D, Bierling M, Speck FD, Mayrhofer KJJ, et al. Fabrication of a robust PEM water electrolyzer based on non-noble metal cathode catalyst: [Mo 3 S 13] 2− clusters anchored to N-doped carbon nanotubes. Small 2020;16. https://doi.org/10.1002/smll.202003161.
[142] Yu H, Danilovic N, Wang Y, Willis W, Poozhikunnath A, Bonville L, et al. Nano-size IrOx catalyst of high activity and stability in PEM water electrolyzer with ultra-low iridium loading. Appl Catal B Environ 2018;239:133–46. https://doi.org/10.1016/j.apcatb.2018.07.064.
[143] Wang S, Lv H, Sun Y, Ji W, Shen X, Zhang C. Constructing supports–network with N–TiO2 nanofibres for highly efficient hydrogen–production of PEM electrolyzer. World Electr Veh J 2021;12:124. https://doi.org/10.3390/wevj12030124.
[144] Mandal M, Moore M, Secanell M. Measurement of the protonic and electronic conductivities of PEM water electrolyzer electrodes. ACS Appl Mater Interfaces 2020;12:49549–62. https://doi.org/10.1021/acsami.0c12111.
[145] Sun CY, Cong YP, Zhang QD, Fu MW, Li L. Element diffusion model with variable coefficient in bimetallic bonding process. J Mater Process Technol 2018;253:99–108. https://doi.org/10.1016/j.jmatprotec.2017.10.045.
[146] Muntean R, Pascal D-T, Rost U, Mărginean G, Brodmann M, Vaszilcsin N. Synthesis and characterisation of platinum–cobalt–manganese ternary alloy catalysts supported on carbon nanofibers: an alternative catalyst for hydrogen evolution reaction. Int J Hydrogen Energy 2020;45:26217–25. https://doi.org/10.1016/j.ijhydene.2020.02.041.
[147] Rakočević L, Srejić I, Maksić A, Golubović J, Štrbac S. Hydrogen evolution on reduced graphene oxide-supported PdAu nanoparticles. Catalysts 2021;11:481. https://doi.org/10.3390/catal11040481.
[148] Tajuddin AAH, Elumalai G, Xi Z, Hu K, Jeong S, Nagasawa K, et al. Corrosion-resistant non-noble metal electrodes for PEM-type water electrolyzer. Int J Hydrogen Energy 2021;46:38603–11. https://doi.org/10.1016/j.ijhydene.2021.09.116.
[149] Xie Z, Yu S, Ma X, Li K, Ding L, Wang W, et al. MoS2 nanosheet integrated electrodes with engineered 1T-2H phases and defects for efficient hydrogen production in practical PEM electrolysis. Appl Catal B Environ 2022;313:121458. https://doi.org/10.1016/j.apcatb.2022.121458.
[150] Dharman RK, Mary Francis B, Ponraj JS, Dhanabalan SC, Manavalan RK, Oh TH. Study on Nickel-induced 1T/2H MoS2 nanostructures in realizing efficient electrocatalysts for hydrogen evolution reaction. J Electroanal Chem 2022;925:116905. https://doi.org/10.1016/j.jelechem.2022.116905.
[151] Pérez G, Díaz-Sainz G, Gómez-Coma L, Álvarez-Miguel L, Garnier A, Cabon N, et al. Rhodium-based cathodes with ultra-low metal loading to increase the sustainability in the hydrogen evolution reaction. J Environ Chem Eng 2022;10:107682. https://doi.org/10.1016/j.jece.2022.107682.
[152] Huang X, Lu R, Cen Y, Wang D, Jin S, Chen W, et al. Micropore-confined Ru nanoclusters catalyst for efficient pH-universal hydrogen evolution reaction. Nano Res 2023;16:9073–80. https://doi.org/10.1007/s12274-023-5711-1.
[153] Hao S, Liu M, Pan J, Liu X, Tan X, Xu N, et al. Dopants fixation of Ruthenium for boosting acidic oxygen evolution stability and activity. Nat Commun 2020;11:5368. https://doi.org/10.1038/s41467-020-19212-y.
[154] Min X, Shi Y, Lu Z, Shen L, Ogundipe TO, Gupta P, et al. High performance and cost-effective supported IrOx catalyst for proton exchange membrane water electrolysis. Electrochim Acta 2021;385:138391. https://doi.org/10.1016/j.electacta.2021.138391.
[155] Zhang J, Fu X, Xia F, Zhang W, Ma D, Zhou Y, et al. Core-shell nanostructured Ru@Ir–O electrocatalysts for superb oxygen evolution in acid. Small 2022;18. https://doi.org/10.1002/smll.202108031.
[156] Senthil Raja D, Cheng P-Y, Cheng C-C, Chang S-Q, Huang C-L, Lu S-Y. In-situ grown metal-organic framework-derived carbon-coated Fe-doped cobalt oxide nanocomposite on fluorine-doped tin oxide glass for acidic oxygen evolution reaction. Appl Catal B Environ 2022;303:120899. https://doi.org/10.1016/j.apcatb.2021.120899.

[157] Joo J, Park Y, Kim J, Kwon T, Jun M, Ahn D, et al. Mn-Dopant differentiating the Ru and Ir oxidation states in catalytic oxides toward durable oxygen evolution reaction in acidic electrolyte. Small Methods 2022;6. https://doi.org/10.1002/smtd.202101236.
[158] Pham C Van, Bühler M, Knöppel J, Bierling M, Seeberger D, Escalera-López D, et al. IrO2 coated TiO2 core-shell microparticles advance performance of low loading proton exchange membrane water electrolyzers. Appl Catal B Environ 2020;269:118762. https://doi.org/10.1016/j.apcatb.2020.118762.
[159] Wang Y, Pang Y, Xu H, Martinez A, Chen KS. PEM Fuel cell and electrolysis cell technologies and hydrogen infrastructure development – a review. Energy Environ Sci 2022;15:2288–328. https://doi.org/10.1039/D2EE00790H.
[160] Wu Q, Wang Y, Zhang K, Xie Z, Sun K, An W, et al. Advances and status of anode catalysts for proton exchange membrane water electrolysis technology. Mater Chem Front 2023;7:1025–45. https://doi.org/10.1039/D3QM00010A.
[161] Huang T, Bian Z-N, Wei C, Huang T, Wang Y-F, Liu Z-H, et al. Ruthenium-iridium alloyed oxides with remarkable catalytic stability for proton exchange membrane water electrolysis at industrial current density. Rare Met 2024. https://doi.org/10.1007/s12598-024-03004-0.
[162] Shi Z, Li J, Wang Y, Liu S, Zhu J, Yang J, et al. Customized reaction route for ruthenium oxide towards stabilized water oxidation in high-performance PEM electrolyzers. Nat Commun 2023;14:843. https://doi.org/10.1038/s41467-023-36380-9.
[163] Doan TL, Lee HE, Shah SSH, Kim M, Kim C, Cho H, et al. A review of the porous transport layer in polymer electrolyte membrane water electrolysis. Int J Energy Res 2021;45:14207–20. https://doi.org/10.1002/er.6739.
[164] Leonard E, Shum AD, Danilovic N, Capuano C, Ayers KE, Pant LM, et al. Interfacial analysis of a PEM electrolyzer using X-ray computed tomography. Sustain Energy Fuels 2020;4:921–31. https://doi.org/10.1039/C9SE00364A.
[165] Schuler T, Ciccone JM, Krentscher B, Marone F, Peter C, Schmidt TJ, et al. Hierarchically structured porous transport layers for polymer electrolyte water electrolysis. Adv Energy Mater 2020;10. https://doi.org/10.1002/aenm.201903216.
[166] Panchenko O, Borgardt E, Zwaygardt W, Hackemüller FJ, Bram M, Kardjilov N, et al. In-situ two-phase flow investigation of different porous transport layer for a polymer electrolyte membrane (PEM) electrolyzer with neutron spectroscopy. J Power Sources 2018;390:108–15. https://doi.org/10.1016/j.jpowsour.2018.04.044.
[167] Kim PJ, Lee JK, Lee C, Fahy KF, Shrestha P, Krause K, et al. Tailoring catalyst layer interface with titanium mesh porous transport layers. Electrochim Acta 2021;373:137879. https://doi.org/10.1016/j.electacta.2021.137879.
[168] Seweryn J, Biesdorf J, Schmidt TJ, Boillat P. Communication—neutron radiography of the water/gas distribution in the porous layers of an operating electrolyser. J Electrochem Soc 2016;163:F3009–11. https://doi.org/10.1149/2.0641607jes.
[169] Bromberger K, Ghinaiya J, Lickert T, Fallisch A, Smolinka T. Hydraulic ex situ through-plane characterization of porous transport layers in PEM water electrolysis cells. Int J Hydrogen Energy 2018;43:2556–69. https://doi.org/10.1016/j.ijhydene.2017.12.042.
[170] Pushkarev AS, Pushkareva IV, Solovyev MA, Prokop M, Bystron T, Rajagopalan SK, et al. On the influence of porous transport layers parameters on the performances of polymer electrolyte membrane water electrolysis cells. Electrochim Acta 2021;399:139436. https://doi.org/10.1016/j.electacta.2021.139436.
[171] Zhao B, Lee C, Lee JK, Fahy KF, LaManna JM, Baltic E, et al. Superhydrophilic porous transport layer enhances efficiency of polymer electrolyte membrane electrolyzers. Cell Reports Phys Sci 2021;2:100580. https://doi.org/10.1016/j.xcrp.2021.100580.
[172] Parra-Restrepo J, Bligny R, Dillet J, Didierjean S, Stemmelen D, Moyne C, et al. Influence of the porous transport layer properties on the mass and charge transfer in a segmented PEM electrolyzer. Int J Hydrogen Energy 2020;45:8094–106. https://doi.org/10.1016/j.ijhydene.2020.01.100.
[173] Wrubel JA, Kang Z, Witteman L, Zhang F-Y, Ma Z, Bender G. Mathematical modeling of novel porous transport layer architectures for proton exchange membrane electrolysis cells. Int J Hydrogen Energy 2021;46:25341–54. https://doi.org/10.1016/j.ijhydene.2021.05.070.
[174] Chen Q, Wang Y, Yang F, Xu H. Two-dimensional multi-physics modeling of porous transport layer in polymer electrolyte membrane electrolyzer for water splitting. Int J Hydrogen Energy 2020;45:32984–94. https://doi.org/10.1016/j.ijhydene.2020.09.148.
[175] Lv H, Chen J, Zhou W, Shen X, Zhang C. Mechanisms analyses and optimization strategies for performance improvement in low-temperature water electrolysis systems via the perspective of mass transfer: a review. Renew Sustain Energy Rev 2023;183:113394. https://doi.org/10.1016/j.rser.2023.113394.
[176] Teuku H, Alshami I, Goh J, Masdar MS, Loh KS. Review on bipolar plates for low-temperature polymer electrolyte membrane water electrolyzer. Int J Energy Res 2021;45:20583–600. https://doi.org/10.1002/er.7182.
[177] Wilhelm Sievers G, Anklam K, Henkel R, Hickmann T, Brüser V. Corrosion-protection of moulded graphite conductive plastic bipolar plates in PEM electrolysis by plasma processing. Int J Hydrogen Energy 2019;44:2435–45. https://doi.org/10.1016/j.ijhydene.2018.12.020.
[178] Liu T, Tao Y, Wang Y, Hu M, Zhang Z, Shao J. Towards cost-effective and durable bipolar plates for proton exchange membrane electrolyzers: a review. Fuel 2024;368:131610. https://doi.org/10.1016/j.fuel.2024.131610.
[179] Kellenberger A, Duca D, Vaszilcsin N, Craciunescu CM. Electrochemical evaluation of niobium corrosion resistance in simulated anodic PEM electrolyzer environment. Int J Electrochem Sci 2020;15:10664–73. https://doi.org/10.20964/2020.11.47.
[180] Sánchez-Molina M, Amores E, Rojas N, Kunowsky M. Additive manufacturing of bipolar plates for hydrogen production in proton exchange membrane water electrolysis cells. Int J Hydrogen Energy 2021;46:38983–91. https://doi.org/10.1016/j.ijhydene.2021.09.152.
[181] Rojas N, Sánchez-Molina M, Sevilla G, Amores E, Almandoz E, Esparza J, et al. Coated stainless steels evaluation for bipolar plates in PEM water electrolysis conditions. Int J Hydrogen Energy 2021;46:25929–43. https://doi.org/10.1016/j.ijhydene.2021.03.100.
[182] Taner T, Naqvi SAH, Ozkaymak M. Techno-economic analysis of a more efficient hydrogen generation system prototype: a case study of PEM electrolyzer with Cr-C coated SS304 bipolar plates. Fuel Cells 2019;19:19–26. https://doi.org/10.1002/fuce.201700225.
[183] Yang G, Xie Z, Yu S, Li K, Li Y, Ding L, et al. All-in-one bipolar electrode: a new concept for compact and efficient water electrolyzers. Nano Energy 2021;90:106551. https://doi.org/10.1016/j.nanoen.2021.106551.
[184] Ye H, Tu Z, Li S. Electrochemical performance of metal nitride coated titanium bipolar plate for proton exchange membrane water electrolyser. J Power Sources 2024;595:234052. https://doi.org/10.1016/j.jpowsour.2024.234052.
[185] Cheng H, Luo H, Wang X, Pan Z, Zhao Q, Dong C, et al. Improving the performance of titanium bipolar plate in proton exchange membrane water electrolysis environment by nitrogen-chromium composite cathode plasma electrolytic deposition. Int J Hydrogen Energy 2023;48:38557–68. https://doi.org/10.1016/j.ijhydene.2023.06.177.
[186] Liu G, Shan D, Fang B, Wang X. Novel hybrid coating of TiN and carbon with improved corrosion resistance for bipolar plates of PEM water electrolysis. Int J Hydrogen Energy 2023;48:18996–9007. https://doi.org/10.1016/j.ijhydene.2023.02.015.
[187] Lædre S, Kongstein OE, Oedegaard A, Karoliussen H, Seland F. Materials for Proton Exchange Membrane water electrolyzer bipolar plates. Int J Hydrogen Energy 2017;42:2713–23. https://doi.org/10.1016/j.ijhydene.2016.11.106.
[188] Ham K, Bae S, Lee J. Classification and technical target of water electrolysis for hydrogen production. J Energy Chem 2024;95:554–76. https://doi.org/10.1016/j.jechem.2024.04.003.
[189] Fiedler L, Ma TC, Fritsch B, Risse JH, Lechner M, Dworschak D, et al. Stability of bipolar plates in proton exchange membrane water electrolysis: dissolution studies of titanium and stainless steel. ECS Meet Abstr 2024. https://doi.org/10.1149/MA2024-01341900mtgabs. MA2024-01:1900–1900.
[190] Luo X, Ren C, Song J, Luo H, Xiao K, Zhang D, et al. Design and fabrication of bipolar plates for PEM water electrolyser. J Mater Sci Technol 2023;146:19–41. https://doi.org/10.1016/j.jmst.2022.10.039.
[191] Liu G, Hou F, Wang X, Fang B. Conductive polymer and nanoparticle-promoted polymer hybrid coatings for metallic bipolar plates in proton membrane exchange water electrolysis. Appl Sci 2023;13:1244. https://doi.org/10.3390/app13031244.
[192] Stiber S, Hehemann M, Carmo M, Müller M, Ayers KE, Capuano C, et al. Long-term operation of Nb-coated stainless steel bipolar plates for proton exchange membrane water electrolyzers. Adv Energy Sustain Res 2022;3. https://doi.org/10.1002/aesr.202200024.
[193] Kellenberger A, Vaszilcsin N, Duca D, Dan ML, Duteanu N, Stiber S, et al. Towards replacing titanium with copper in the bipolar plates for proton exchange membrane water electrolysis. Materials 2022;15:1628. https://doi.org/10.3390/ma15051628.
[194] Tijani AS, Binti Kamarudin NA, Binti Mazlan FA. Investigation of the effect of charge transfer coefficient (CTC) on the operating voltage of polymer electrolyte membrane (PEM) electrolyzer. Int J Hydrogen Energy 2018;43:9119–32. https://doi.org/10.1016/j.ijhydene.2018.03.111.
[195] Cembalest MJP. Morgan tenth annual energy paper. 2020.
[196] Finance HC of S. Hydrogen for the future: delivering zero-carbon in heavy industry. 2020.
[197] European Commission. A hydrogen strategy for a climate-neutral Europe EN, vol. 53; 2020.
[198] Collins L. Green hydrogen: ITM power's new gigafactory will cut costs of electrolysers by almost 40 %. 2021.
[199] Bullard N. A gigafactory for hydrogen could Be a game- changer. 2021.
[200] IEA. The future of hydrogen – analysis - IEA. Int Energy Agency; 2019. p. 1–203. https://www.iea.org/reports/the-future-of-hydrogen. [Accessed 28 February 2022].
[201] Buttler A, Spliethoff H. Current status of water electrolysis for energy storage, grid balancing and sector coupling via power-to-gas and power-to-liquids: a review. Renew Sustain Energy Rev 2018;82:2440–54. https://doi.org/10.1016/j.rser.2017.09.003.
[202] Research AE. Hydrogen market attractiveness report (HyMAR). 2021.
[203] IEA. Hydrogen projects database. 2020.
[204] Commission ET. Making the hydrogen economy possible: accelerating clean hydrogen in an electrified economy''. 2021.
[205] Sanderson H. Fuel-cell producers jump on new hydrogen 'hype cycle. ' Financ Times; 2020.
[206] IEA. Global EV outlook. 2021.
[207] Council H. Path to hydrogen competitiveness: a cost perspective. 2020.
[208] Cozzolino R, Bella G. A review of electrolyzer-based systems providing grid ancillary services: current status, market, challenges and future directions. Front Energy Res 2024;12. https://doi.org/10.3389/fenrg.2024.1358333.
[209] CaFCP road map for procurement of fuel cell buses in California. Fuel Cells Bull 2019;2019:10–1. https://doi.org/10.1016/S1464-2859(19)30425-0.

[210] Green I. Hydrogen cost reduction: scaling up electrolysers to meet the 1.5 oC climate goal. 2020.
[211] Bloomberg NEF. Hydrogen economy outlook. Bloom New Energy Financ 2020;12.
[212] Mayyas A, Mann M. Emerging manufacturing technologies for fuel cells and electrolyzers. Procedia Manuf 2019;33:508–15. https://doi.org/10.1016/j.promfg.2019.04.063.
[213] Mayyas A, Mann M. Manufacturing competitiveness analysis for hydrogen refueling stations. Int J Hydrogen Energy 2019;44:9121–42. https://doi.org/10.1016/j.ijhydene.2019.02.135.
[214] Smolinka T, Wiebe N, Thomassen N. Cost break down and cost reduction strategies for PEM water electrolysis systems. 6th Eur PEFC Electrolyser Forum. 2017. p. 1–16.
[215] Glenk G, Reichelstein S. Economics of converting renewable power to hydrogen. Nat Energy 2019;4:216–22. https://doi.org/10.1038/s41560-019-0326-1.
[216] Bertuccioli L, Chan A, Hart D, Lehner F, Madden B, Standen E. Development of water electrolysis in the European union. Fuel Cells Hydrog Jt Undert 2014;1:1–160.
[217] Böhm H, Goers S, Zauner A. Estimating future costs of power-to-gas – a component-based approach for technological learning. Int J Hydrogen Energy 2019;44:30789–805. https://doi.org/10.1016/j.ijhydene.2019.09.230.
[218] James J. Towards sustainable ammonia production. TNO; 2019.
[219] Graf MD, A. EU-wide innovation support is key to the success of electrolysis manufacturing in Europe. Agora Energiewende; 2019.
[220] Wang CR, Stansberry JM, Mukundan R, Chang H-MJ, Kulkarni D, Park AM, et al. Proton exchange membrane (PEM) water electrolysis: cell-level considerations for gigawatt-scale deployment. Chem Rev 2025;125:1257–302. https://doi.org/10.1021/acs.chemrev.3c00904.
[221] Hassan Q, Hafedh SA, Mohammed HB, Abdulrahman IS, Salman HM, Jaszczur M. A review of hydrogen production from bio-energy, technologies and assessments. Energy Harvest Syst 2024;11. https://doi.org/10.1515/ehs-2022-0117.
[222] Hassan Q, Sameen AZ, Salman HM, Jaszczur M. A roadmap with strategic policy toward green hydrogen production: the case of Iraq. Sustainability 2023;15:5258. https://doi.org/10.3390/su15065258.
[223] Hassan Q, Sameen AZ, Salman HM, Jaszczur M, Al-Jiboory AK. Hydrogen energy future: advancements in storage technologies and implications for sustainability. J Energy Storage 2023;72:108404. https://doi.org/10.1016/j.est.2023.108404.
[224] Zhao J, Patwary AK, Qayyum A, Alharthi M, Bashir F, Mohsin M, et al. The determinants of renewable energy sources for the fueling of green and sustainable economy. Energy 2022;238:122029. https://doi.org/10.1016/j.energy.2021.122029.
[225] Hassan Q, Azzawi IDJ, Sameen AZ, Salman HM. Hydrogen fuel cell vehicles: opportunities and challenges. Sustainability 2023;15:11501. https://doi.org/10.3390/su151511501.
[226] Jaszczur M, Hassan Q, Sameen AZ, Salman HM, Olapade OT, Wieteska S. Massive green hydrogen production using solar and wind energy: comparison between Europe and the Middle East. Energies 2023;16:5445. https://doi.org/10.3390/en16145445.
[227] Hassan Q, Algburi S, Sameen AZ, Salman HM, Jaszczur M. Green hydrogen: a pathway to a sustainable energy future. Int J Hydrogen Energy 2024;50:310–33. https://doi.org/10.1016/j.ijhydene.2023.08.321.
[228] Lagadec MF, Grimaud A. Water electrolysers with closed and open electrochemical systems. Nat Mater 2020;19:1140–50. https://doi.org/10.1038/s41563-020-0788-3.
[229] Padgett E, Bender G, Haug A, Lewinski K, Sun F, Yu H, et al. Catalyst layer resistance and utilization in PEM electrolysis. J Electrochem Soc 2023;170:084512. https://doi.org/10.1149/1945-7111/acee25.
[230] Lyu X, Foster J, Rice R, Padgett E, Creel EB, Li J, et al. Aging gracefully? Investigating iridium oxide ink's impact on microstructure, catalyst/ionomer interface, and PEMWE performance. J Power Sources 2023;581:233503. https://doi.org/10.1016/j.jpowsour.2023.233503.
[231] Lickert T, Fischer S, Young JL, Klose S, Franzetti I, Hahn D, et al. Advances in benchmarking and round robin testing for PEM water electrolysis: reference protocol and hardware. Appl Energy 2023;352:121898. https://doi.org/10.1016/j.apenergy.2023.121898.
[232] Liu J, Liu H, Yang Y, Tao Y, Zhao L, Li S, et al. Efficient and stable proton exchange membrane water electrolysis enabled by stress optimization. ACS Cent Sci 2024. https://doi.org/10.1021/acscentsci.4c00037.
[233] Liu H, Wang X, Lao K, Wen L, Huang M, Liu J, et al. Optimizing ionomer distribution in anode catalyst layer for stable proton exchange membrane water electrolysis. Adv Mater 2024;36. https://doi.org/10.1002/adma.202402780.
[234] Tao HB, Liu H, Lao K, Pan Y, Tao Y, Wen L, et al. The gap between academic research on proton exchange membrane water electrolysers and industrial demands. Nat Nanotechnol 2024;19:1074–6. https://doi.org/10.1038/s41565-024-01699-x.
[235] Jaramillo Rodríguez ND, Luxa A, Jürgensen L. Adaptation and application of a polarisation curve test protocol for a commercial pem electrolyser on cell and stack level. Acta Mech Autom 2023;17:395–404. https://doi.org/10.2478/ama-2023-0045.
[236] Sayed-Ahmed H, Toldy ÁI, Santasalo-Aarnio A. Dynamic operation of proton exchange membrane electrolyzers—critical review. Renew Sustain Energy Rev 2024;189:113883. https://doi.org/10.1016/j.rser.2023.113883.
[237] Wang Q, Cheng Y, Tao HB, Liu Y, Ma X, Li D, et al. Long-term stability challenges and opportunities in acidic oxygen evolution electrocatalysis. Angew Chemie Int Ed 2023;62. https://doi.org/10.1002/anie.202216645.